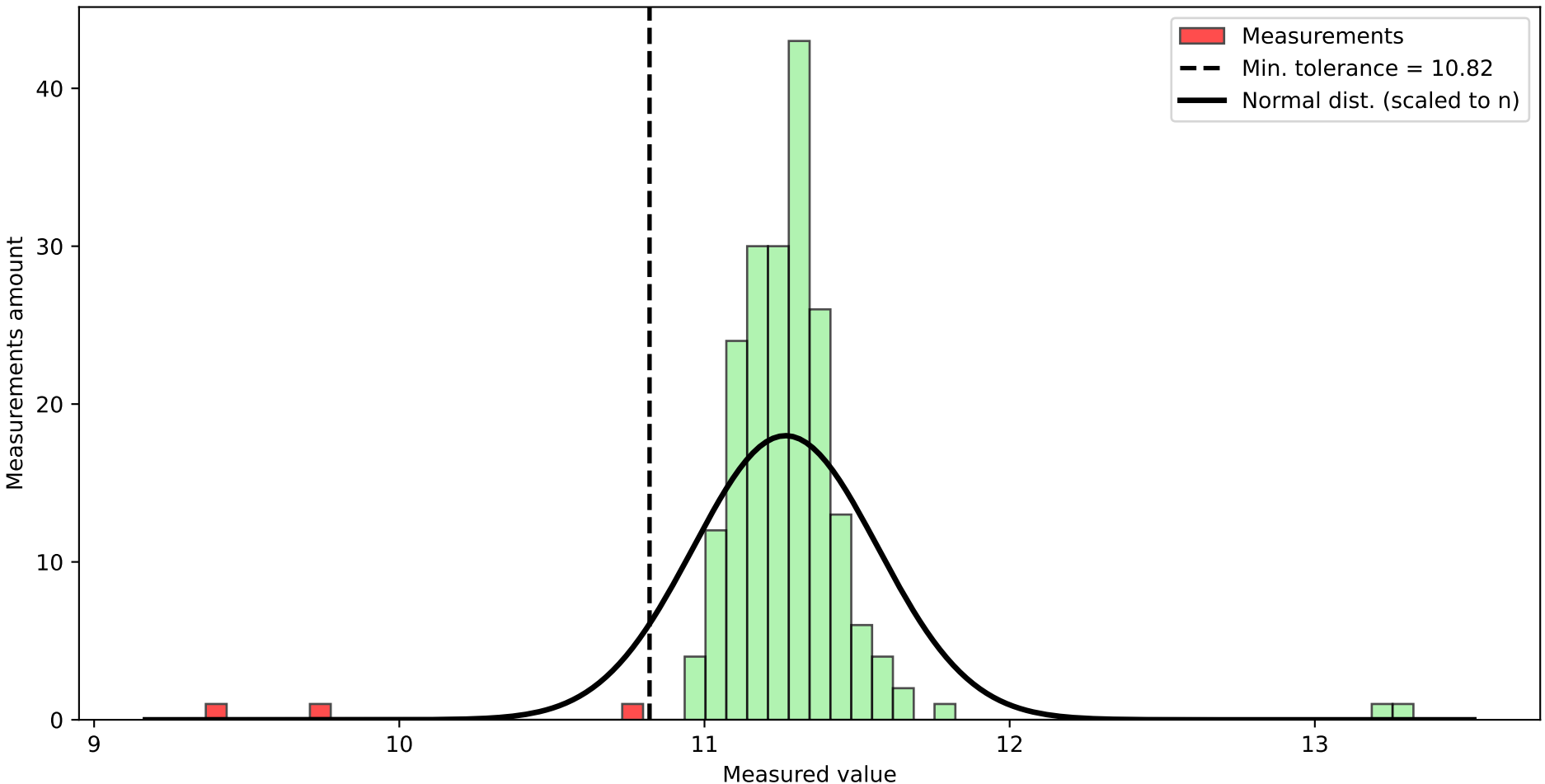
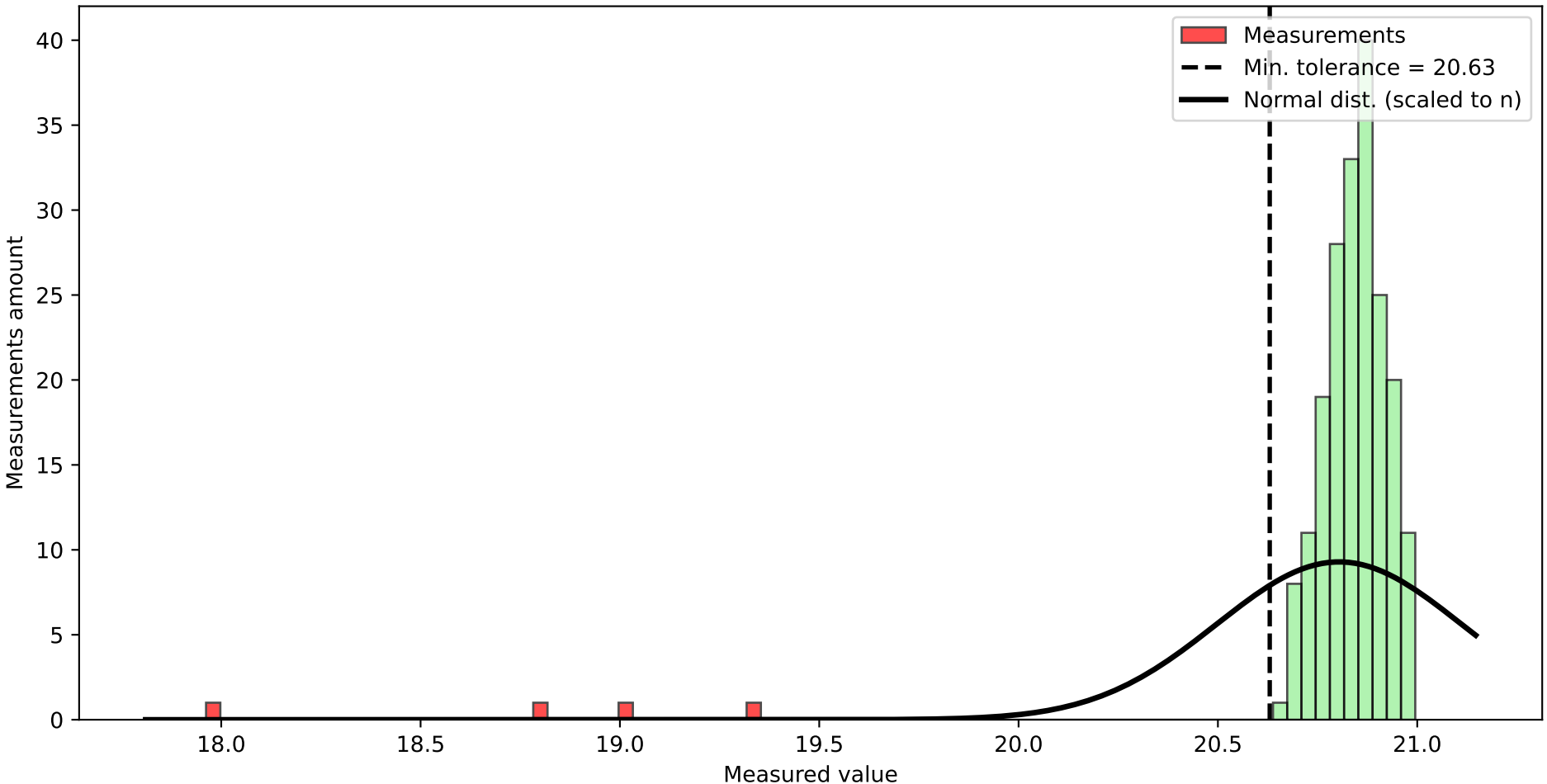
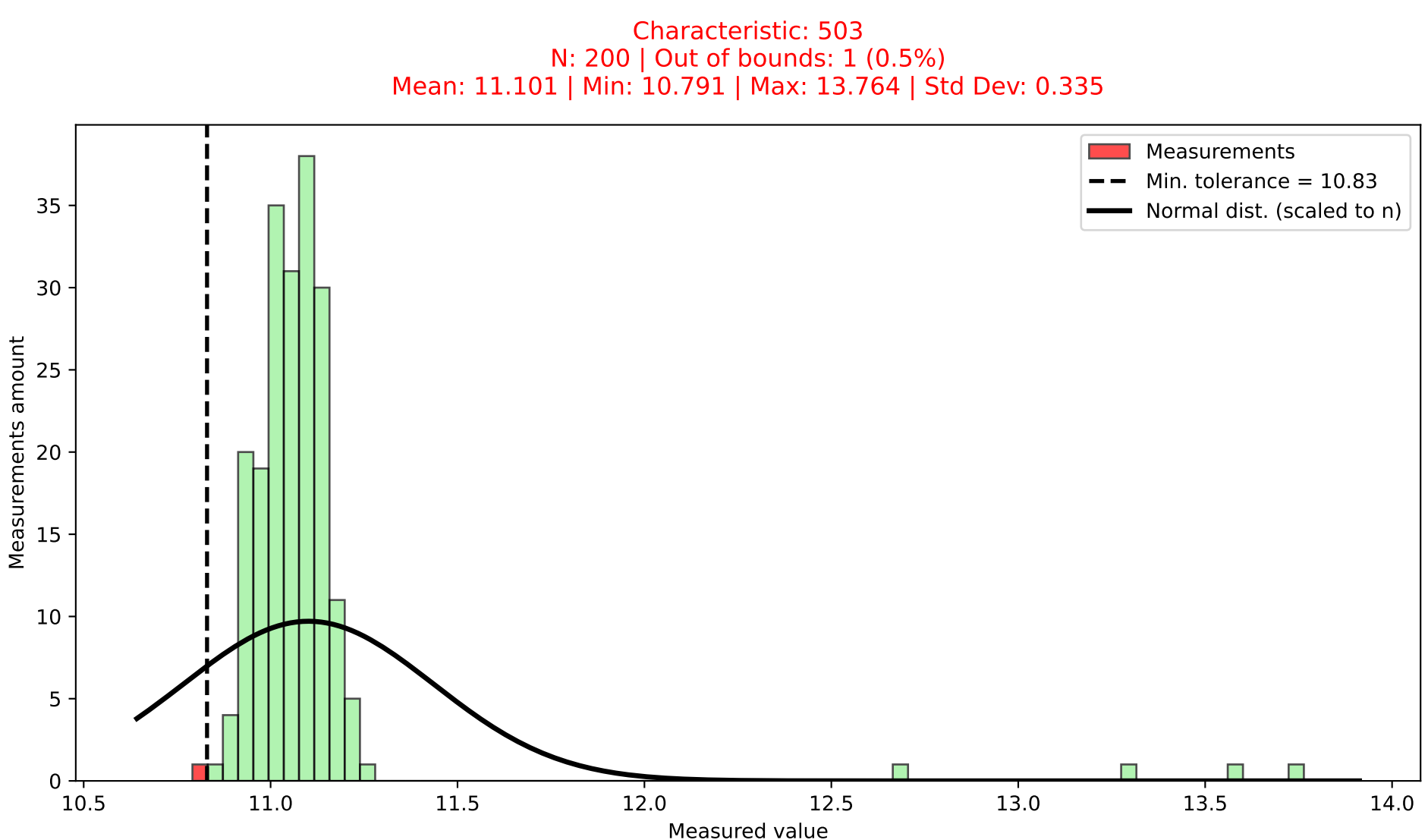
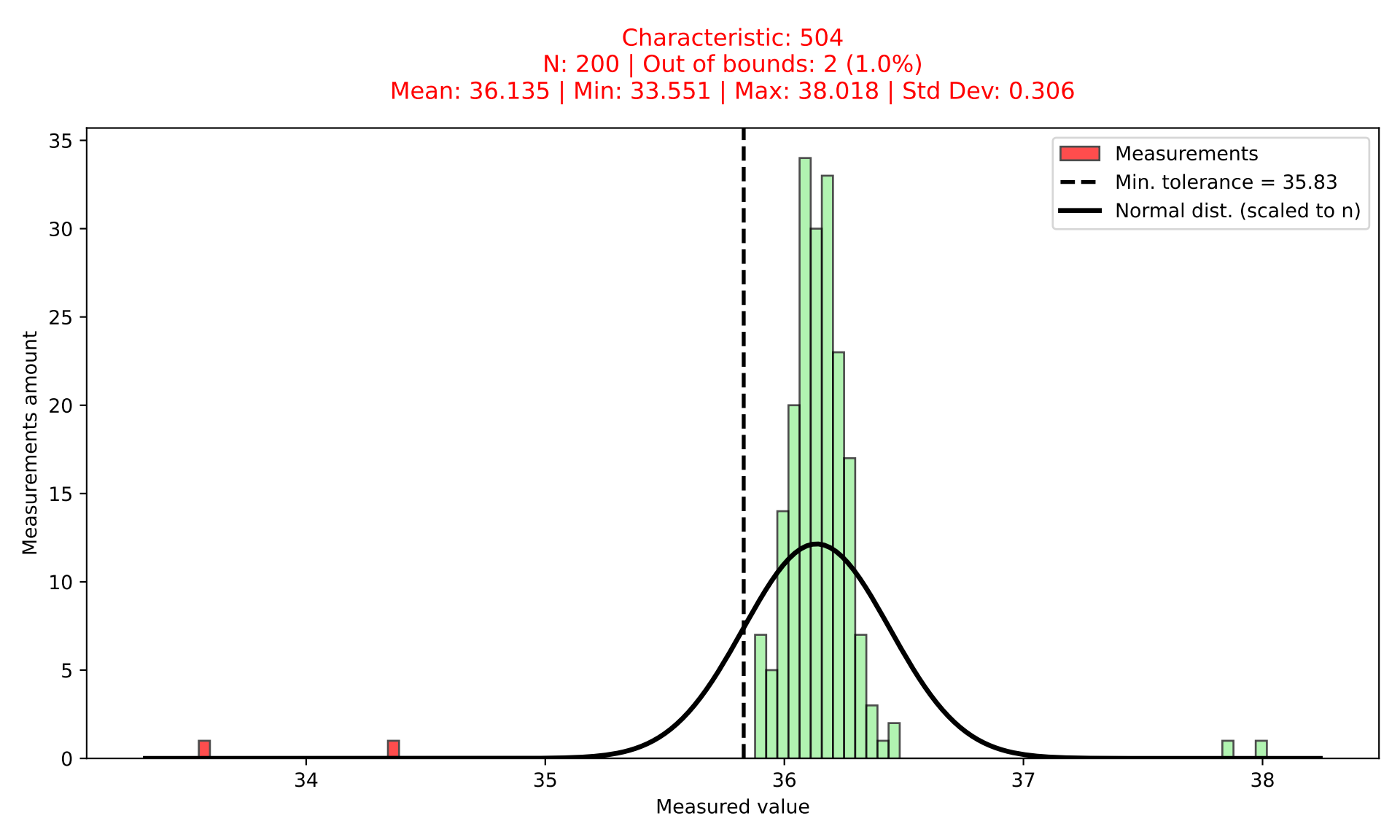
PPAP Dimensional Inspection Summary

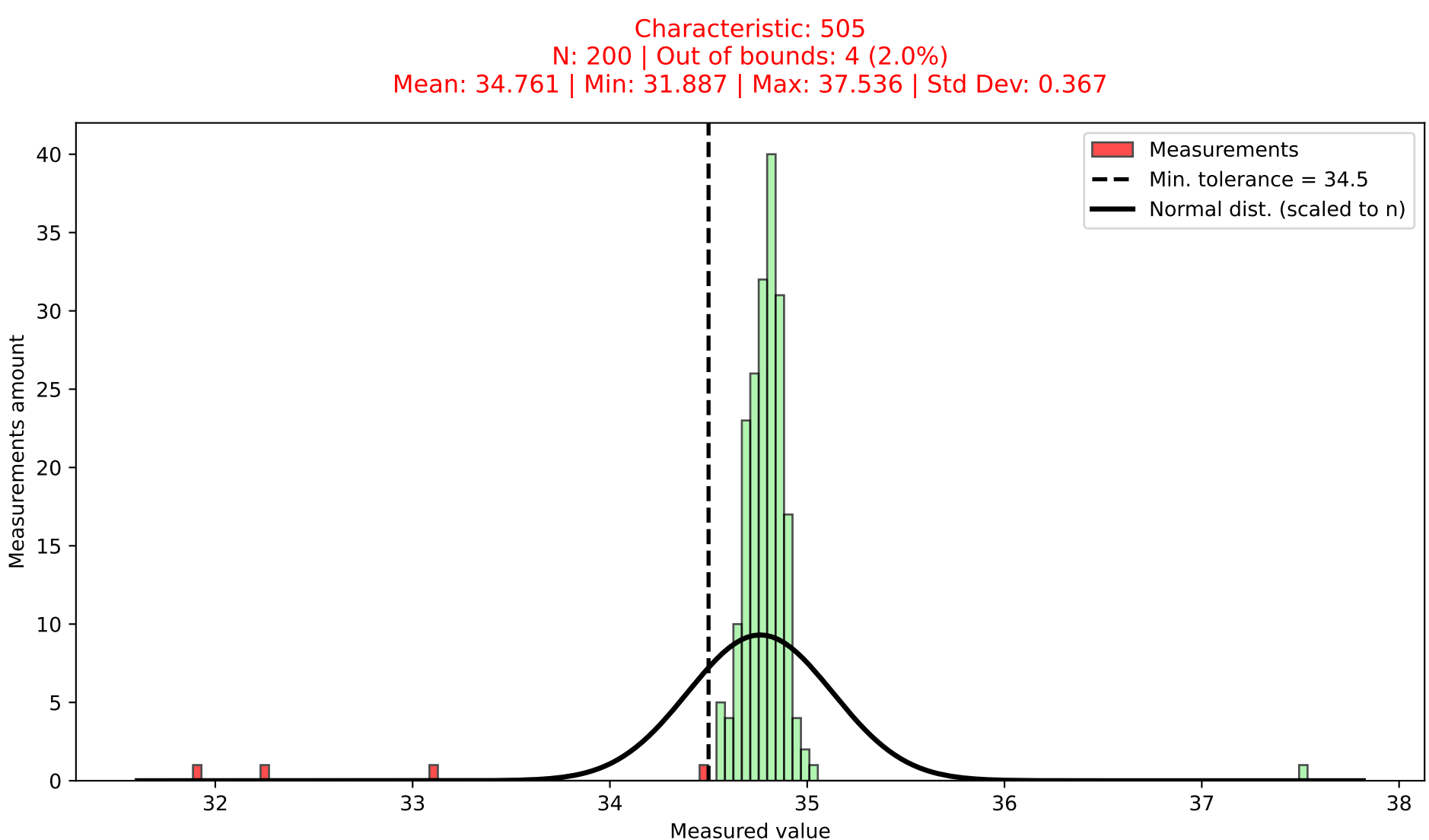
Date: 2026-09-14
Generated by: Python QC Automation Framework

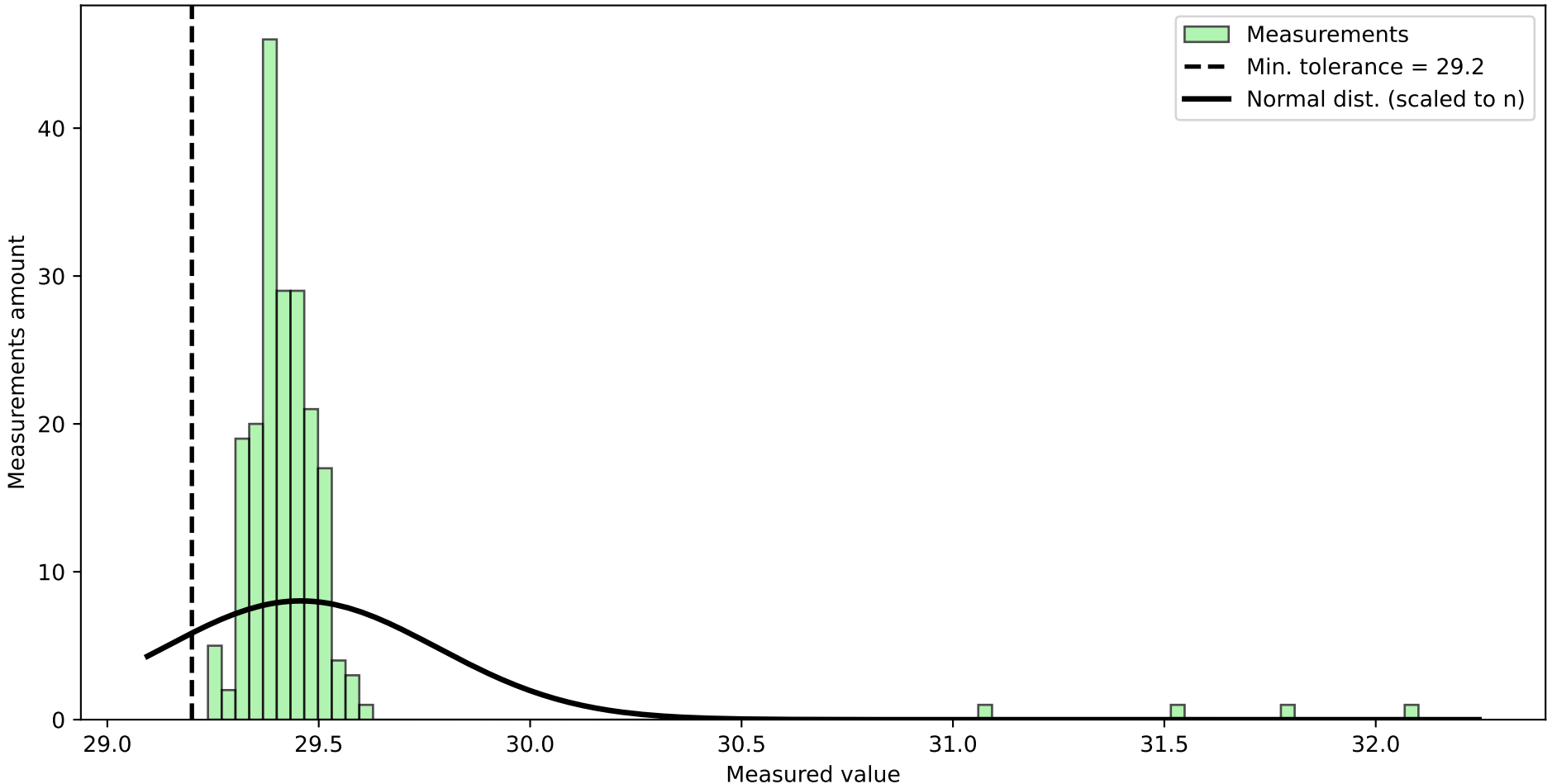

Characteristic: 501
N: 200 | Out of bounds: 3 (1.5%)
Mean: 11.266 | Min: 9.366 | Max: 13.323 | Std Dev: 0.302
Measurements
Min. tolerance = 10.82
Normal dist. (scaled to n)
Measurements amount
Measured value
0
10
20
30
40
9
10
11
12
13

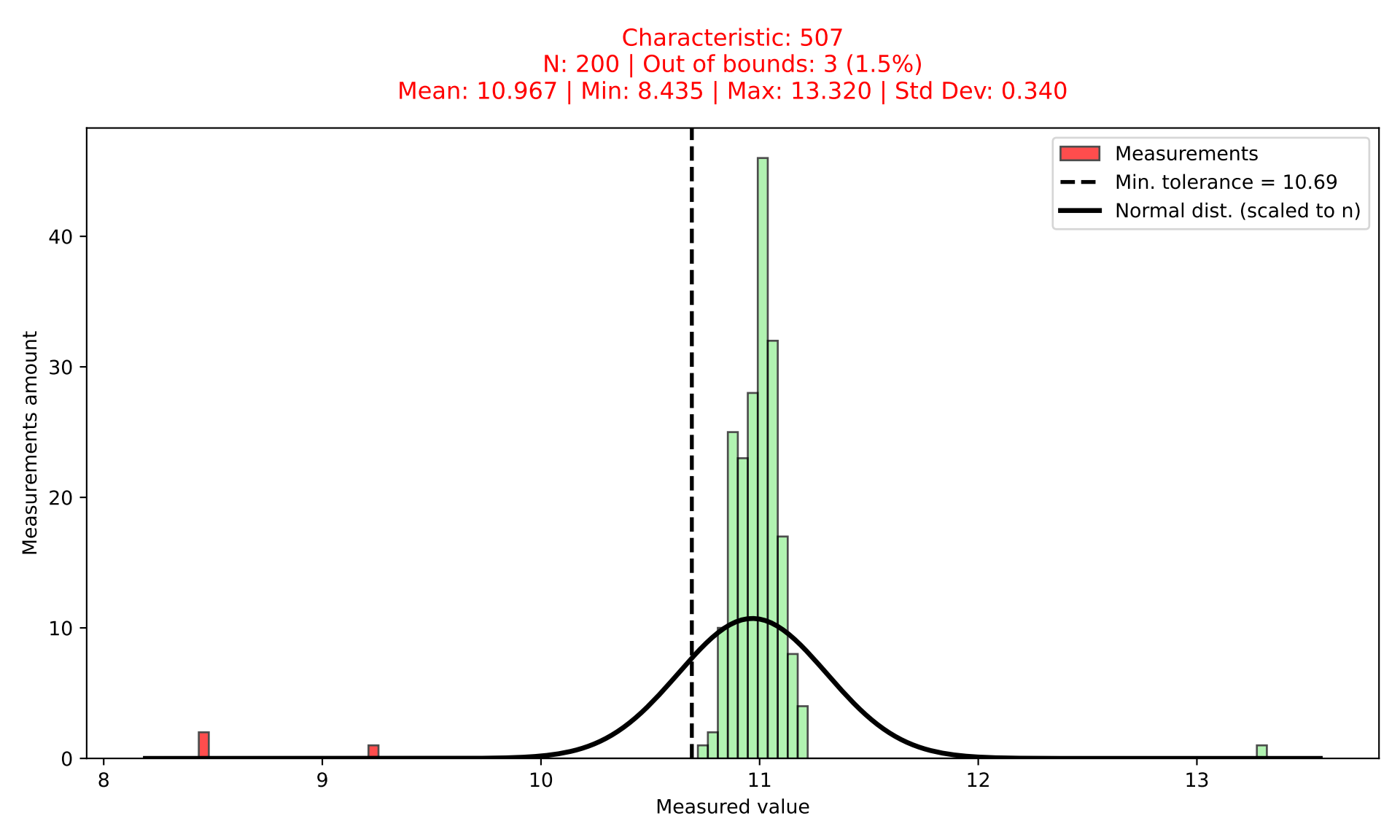

Characteristic: 502
N: 200 | Out of bounds: 4 (2.0%)
Mean: 20.804 | Min: 17.962 | Max: 20.995 | Std Dev: 0.306
Measurements
Min. tolerance = 20.63
Normal dist. (scaled to n)
Measurements amount
Measured value
0
5
10
15
20
25
30
35
40
18.0
18.5
19.0
19.5
20.0
20.5
21.0

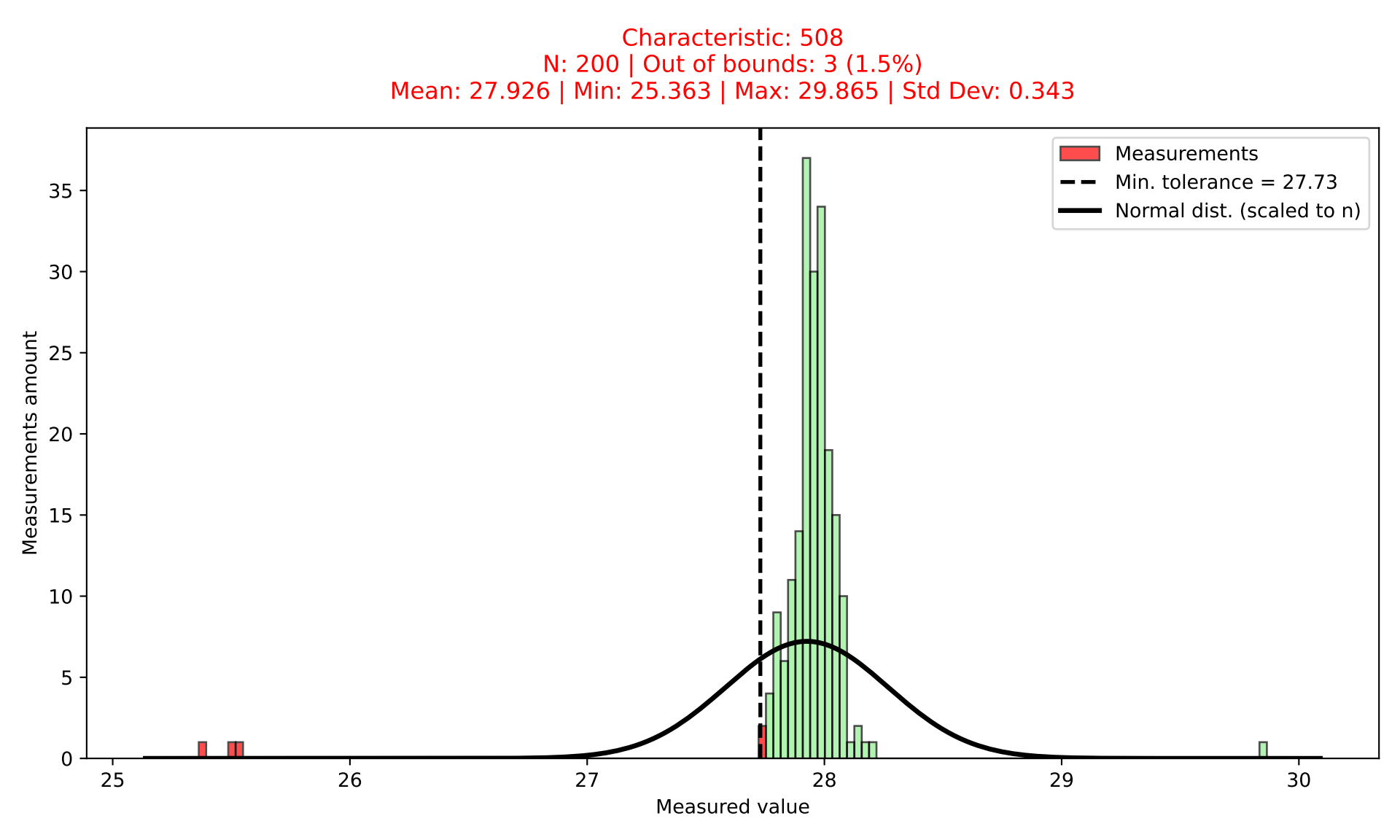

Characteristic: 503
N: 200 | Out of bounds: 1 (0.5%)
Mean: 11.101 | Min: 10.791 | Max: 13.764 | Std Dev: 0.335
Measurements
Min. tolerance = 10.83
Normal dist. (scaled to n)
Measurements amount
Measured value
0
5
10
15
20
25
30
35
10.5
11.0
11.5
12.0
12.5
13.0
13.5
14.0

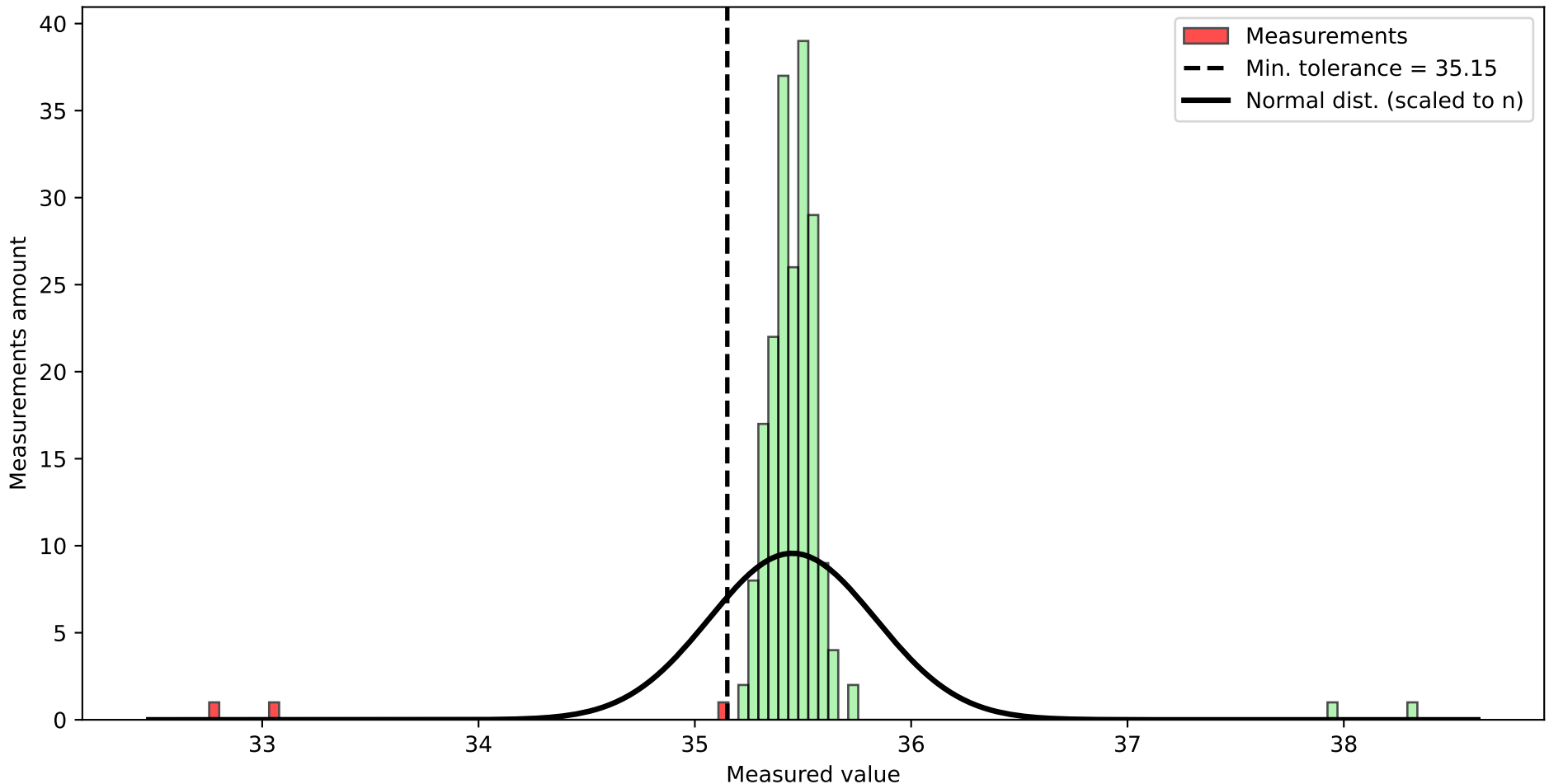

Characteristic: 504
N: 200 | Out of bounds: 2 (1.0%)
Mean: 36.135 | Min: 33.551 | Max: 38.018 | Std Dev: 0.306
Measurements
Min. tolerance = 35.83
Normal dist. (scaled to n)
Measurements amount
Measured value
0
5
10
15
20
25
30
35
34
35
36
37
38

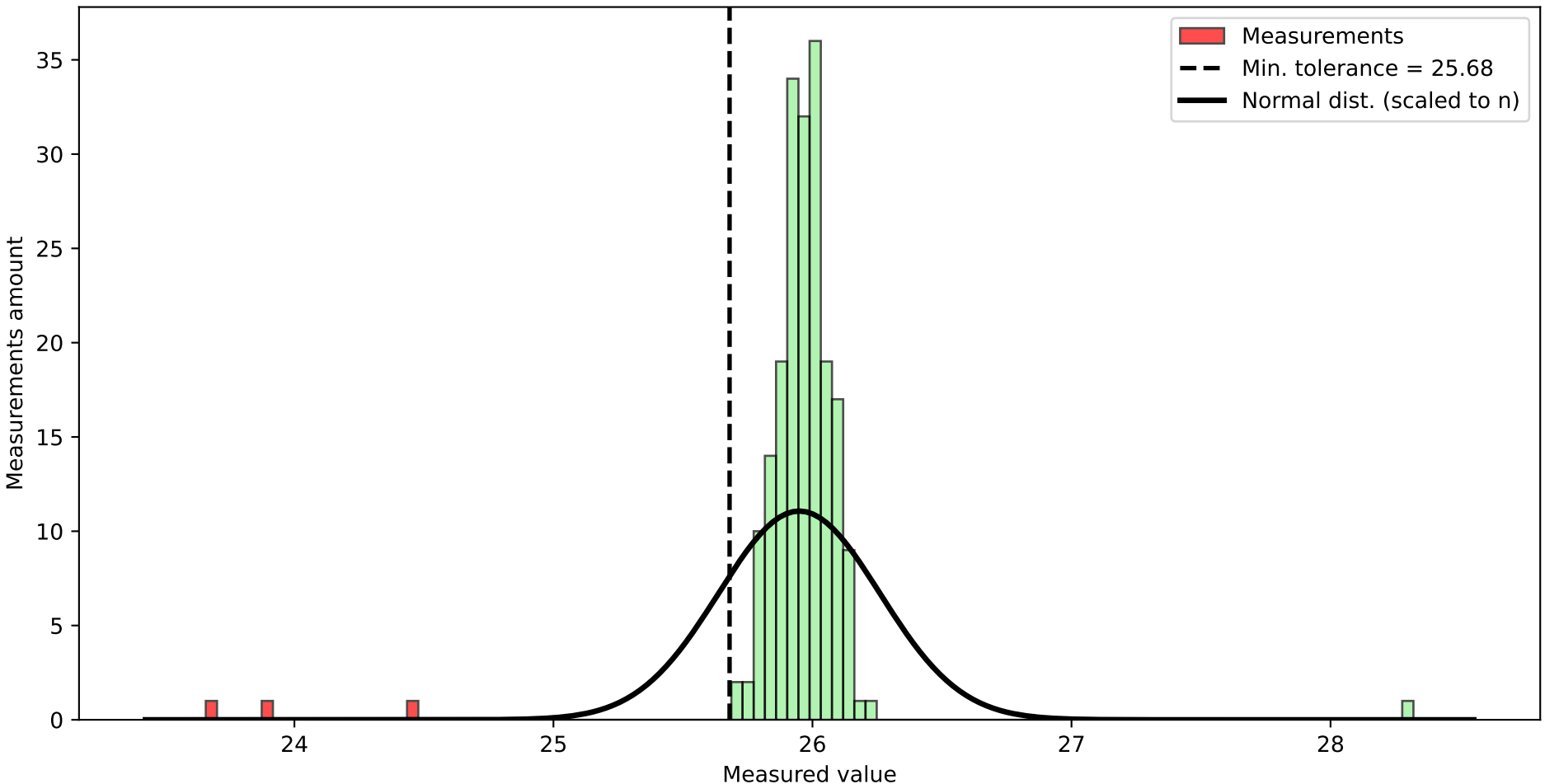

Characteristic: 505
N: 200 | Out of bounds: 4 (2.0%)
Mean: 34.761 | Min: 31.887 | Max: 37.536 | Std Dev: 0.367
Measurements
Min. tolerance = 34.5
Normal dist. (scaled to n)
Measurements amount
Measured value
0
5
10
15
20
25
30
35
40
32
33
34
35
36
37
38


Characteristic: 506
N: 200 | Out of bounds: 0 (0.0%)
Mean: 29.457 | Min: 29.238 | Max: 32.101 | Std Dev: 0.323
Measurements
Min. tolerance = 29.2
Normal dist. (scaled to n)
Measurements amount
Measured value
0
10
20
30
40
29.0
29.5
30.0
30.5
31.0
31.5
32.0

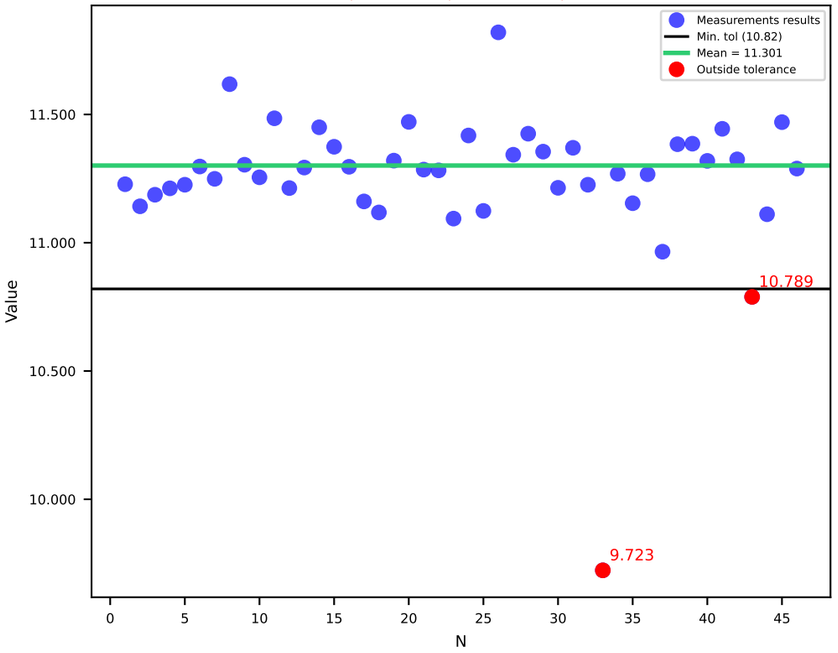

Characteristic: 507
N: 200 | Out of bounds: 3 (1.5%)
Mean: 10.967 | Min: 8.435 | Max: 13.320 | Std Dev: 0.340
Measurements
Min. tolerance = 10.69
Normal dist. (scaled to n)
Measurements amount
Measured value
0
10
20
30
40
8
9
10
11
12
13


Characteristic: 508
N: 200 | Out of bounds: 3 (1.5%)
Mean: 27.926 | Min: 25.363 | Max: 29.865 | Std Dev: 0.343
Measurements
Min. tolerance = 27.73
Normal dist. (scaled to n)
Measurements amount
Measured value
0
5
10
15
20
25
30
35
25
26
27
28
29
30


Characteristic: 509
N: 200 | Out of bounds: 3 (1.5%)
Mean: 35.451 | Min: 32.755 | Max: 38.340 | Std Dev: 0.385
Measurements
Min. tolerance = 35.15
Normal dist. (scaled to n)
Measurements amount
Measured value
0
5
10
15
20
25
30
35
40
33
34
35
36
37
38

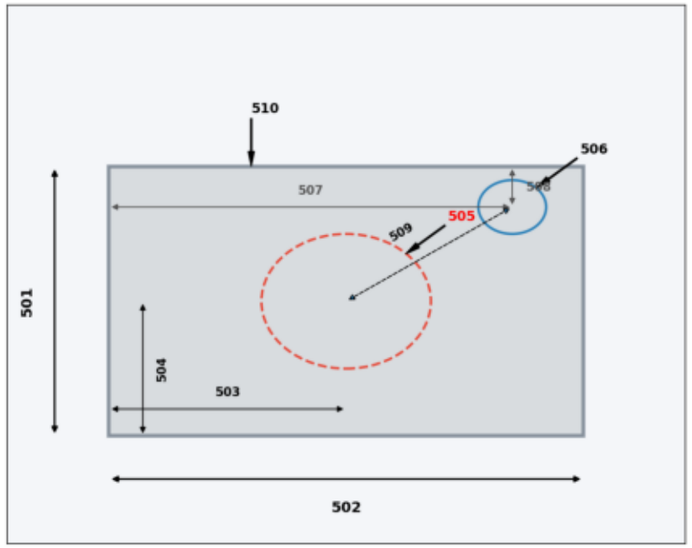

Characteristic: 510
N: 200 | Out of bounds: 3 (1.5%)
Mean: 25.949 | Min: 23.658 | Max: 28.320 | Std Dev: 0.311
Measurements
Min. tolerance = 25.68
Normal dist. (scaled to n)
Measurements amount
0
5
10
15
20
25
30
35
24
25
26
27
28
Measured value

Char: 501 | N: 46 | Out: 2 (4.3%)
Mean: 11.301 | Min: 9.723 | Max: 11.820 | Std Dev: 0.148

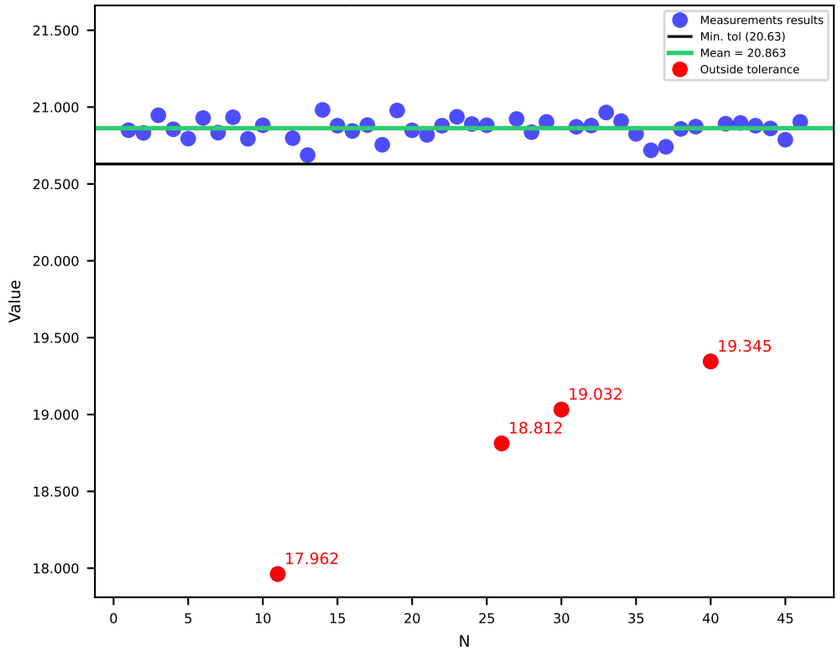


Technical Drawing:

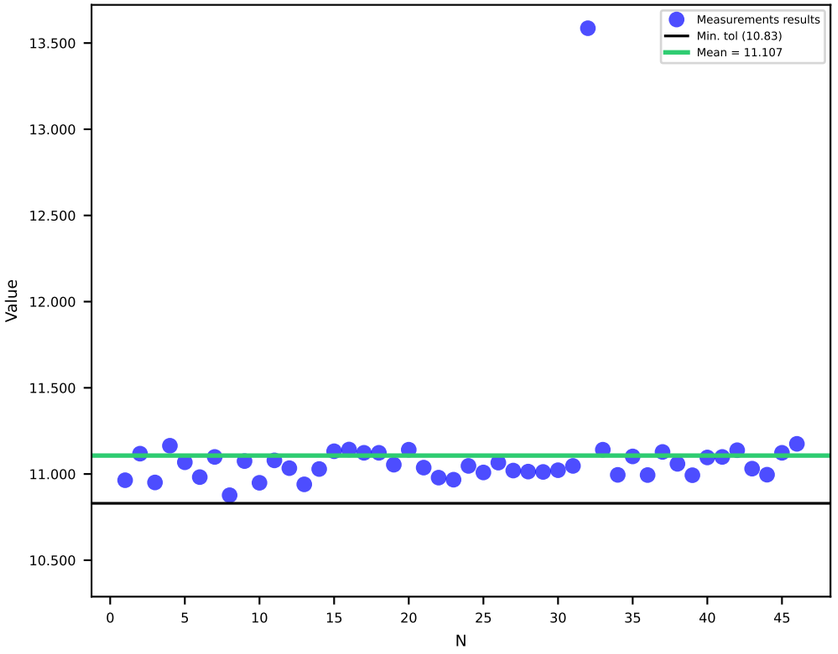
Technical drawing: DETAL-A



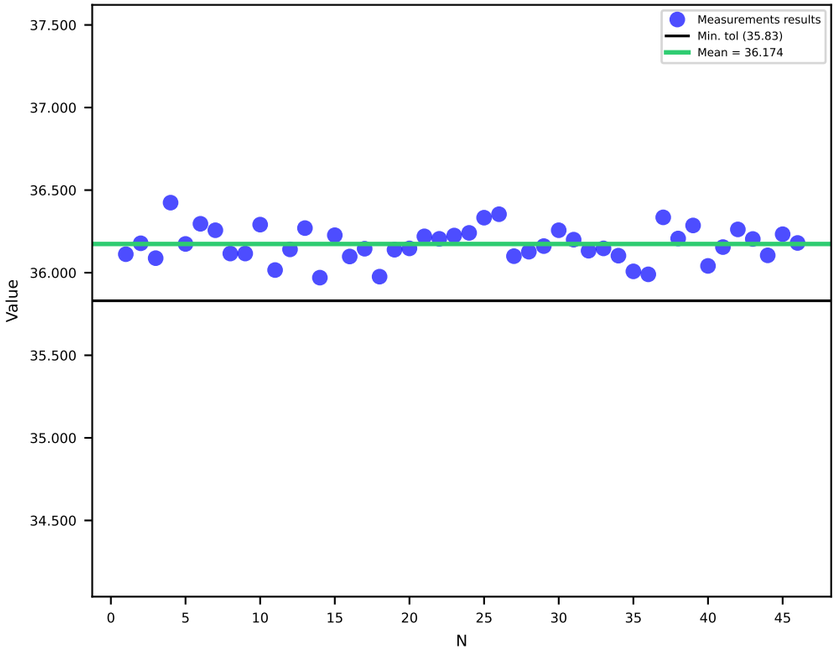
Char: 502 | N: 46 | Out: 4 (8.7%)
Mean: 20.863 | Min: 17.962 | Max: 20.982 | Std Dev: 0.066



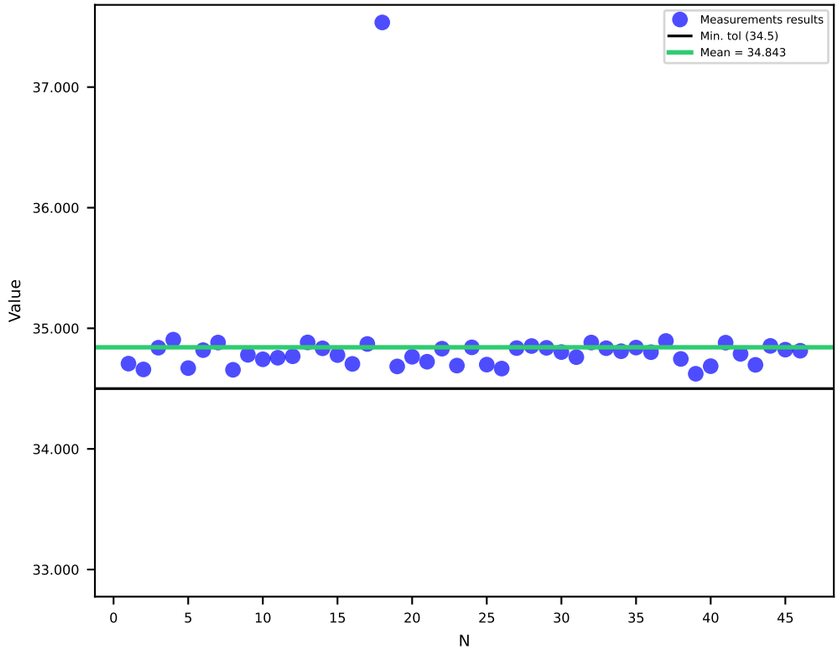
Char: 503 | N: 46 | Out: 0 (0.0%)
Mean: 11.107 | Min: 10.877 | Max: 13.586 | Std Dev: 0.380



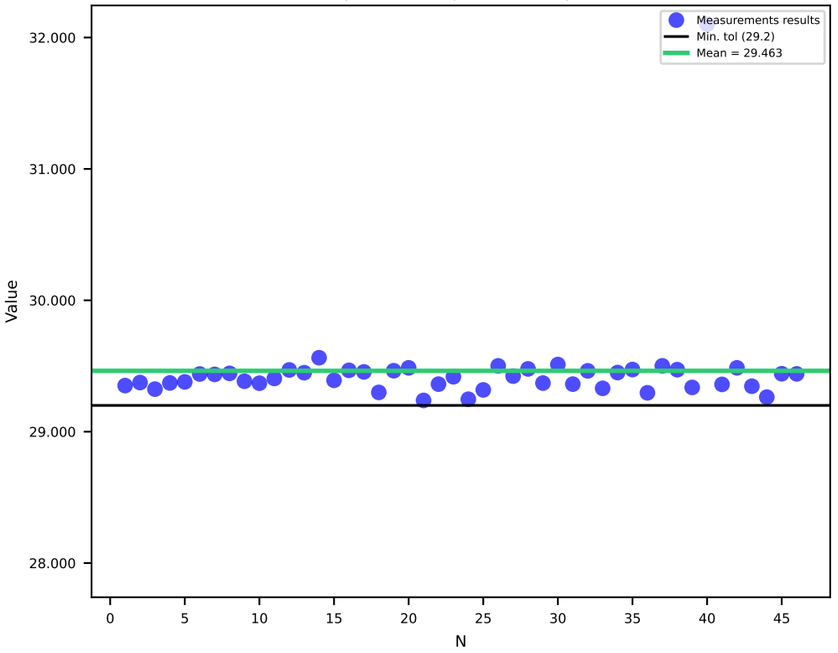
Char: 504 | N: 46 | Out: 0 (0.0%)
Mean: 36.174 | Min: 35.970 | Max: 36.424 | Std Dev: 0.102



Char: 505 | N: 46 | Out: 0 (0.0%)
Mean: 34.843 | Min: 34.623 | Max: 37.536 | Std Dev: 0.413

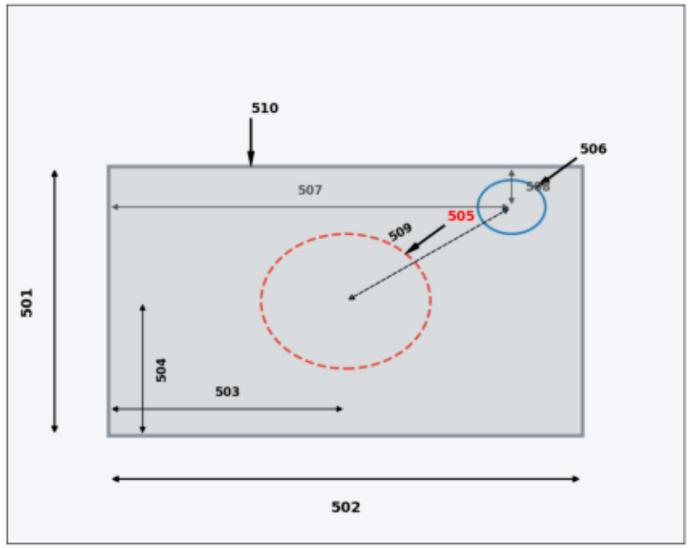


Char: 506 | N: 46 | Out: 0 (0.0%)
Mean: 29.463 | Min: 29.238 | Max: 32.101 | Std Dev: 0.404

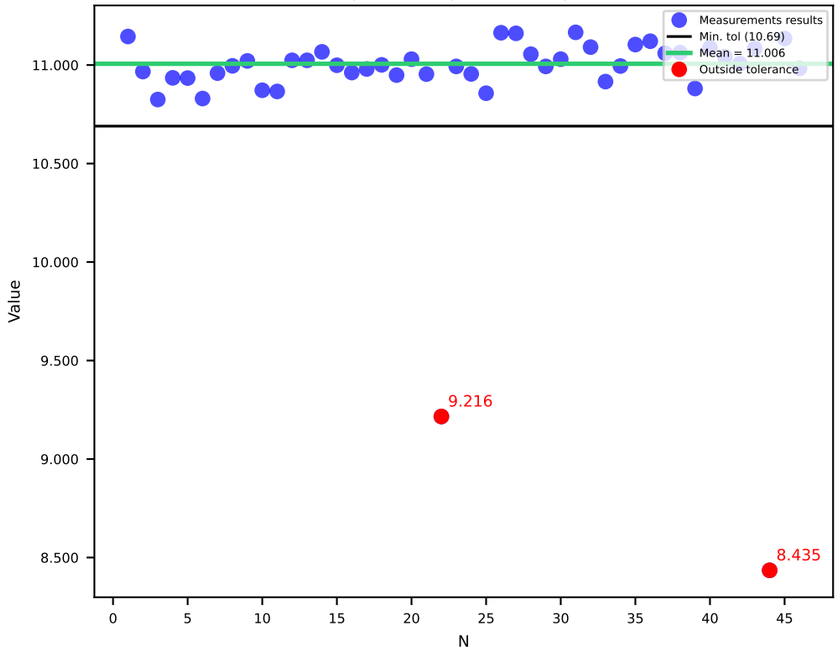


Technical Drawing:

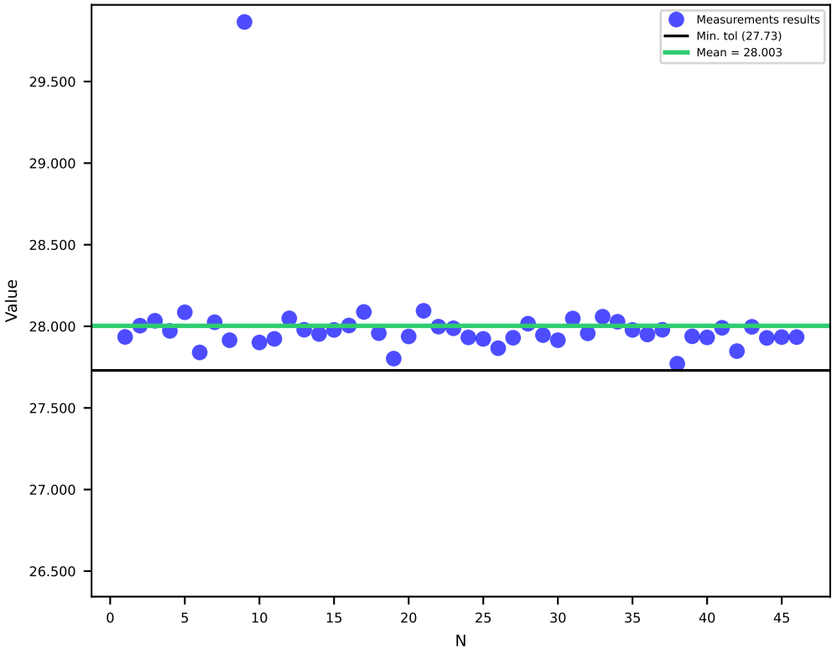
Technical drawing: DETAL-A



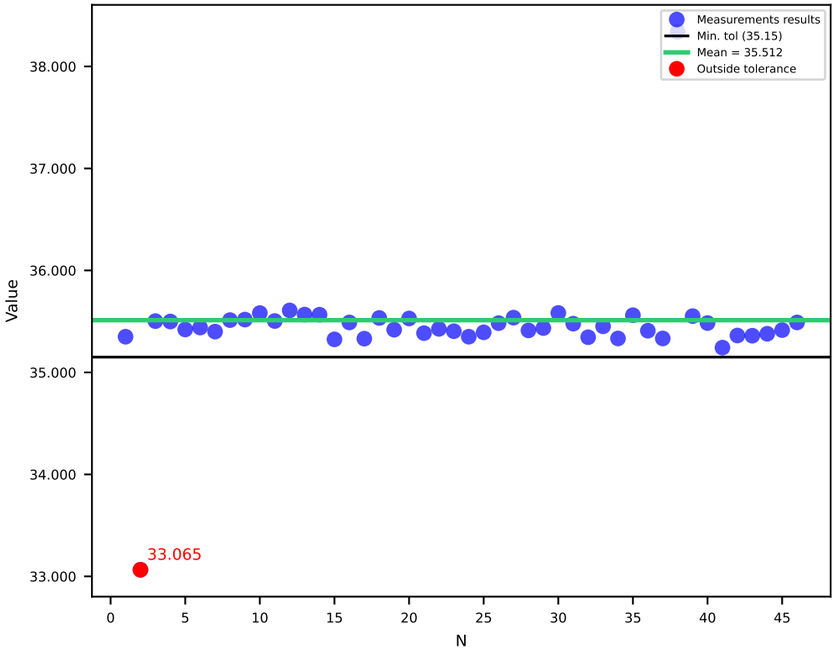
Char: 507 | N: 46 | Out: 2 (4.3%)
Mean: 11.006 | Min: 8.435 | Max: 11.166 | Std Dev: 0.089



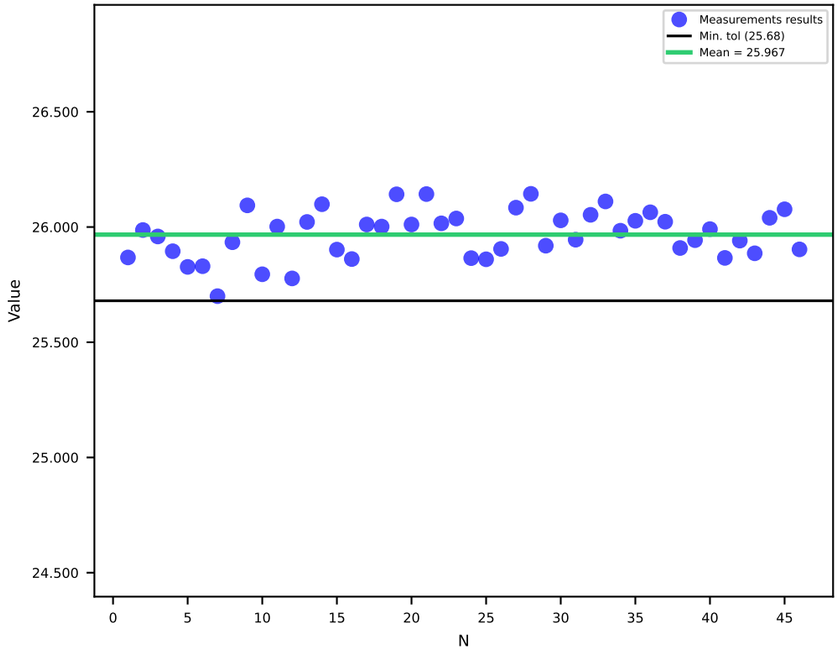
Char: 508 | N: 46 | Out: 0 (0.0%)
Mean: 28.003 | Min: 27.771 | Max: 29.865 | Std Dev: 0.289



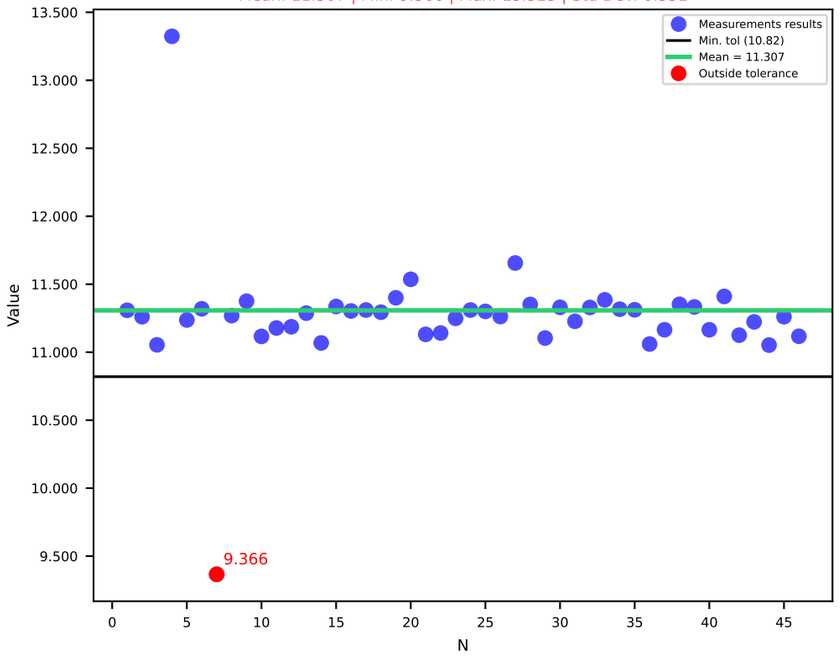
Char: 509 | N: 46 | Out: 1 (2.2%)
Mean: 35.512 | Min: 33.065 | Max: 38.340 | Std Dev: 0.440



Char: 510 | N: 46 | Out: 0 (0.0%)
Mean: 25.967 | Min: 25.700 | Max: 26.144 | Std Dev: 0.104

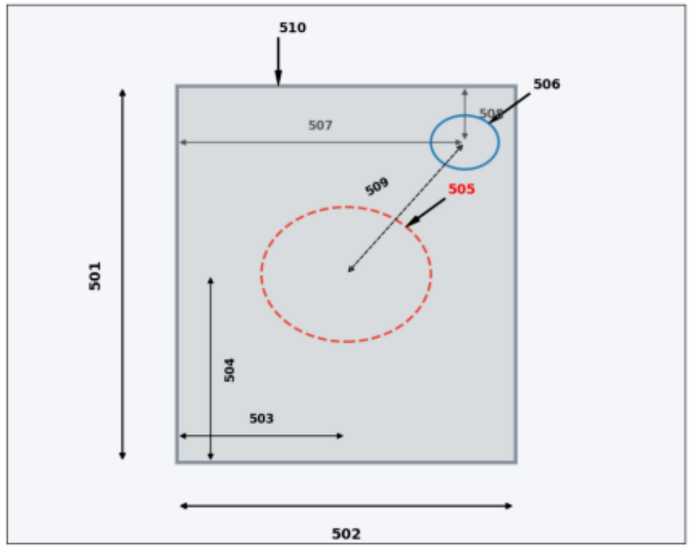


Char: 501 | N: 46 | Out: 1 (2.2%)
Mean: 11.307 | Min: 9.366 | Max: 13.323 | Std Dev: 0.331

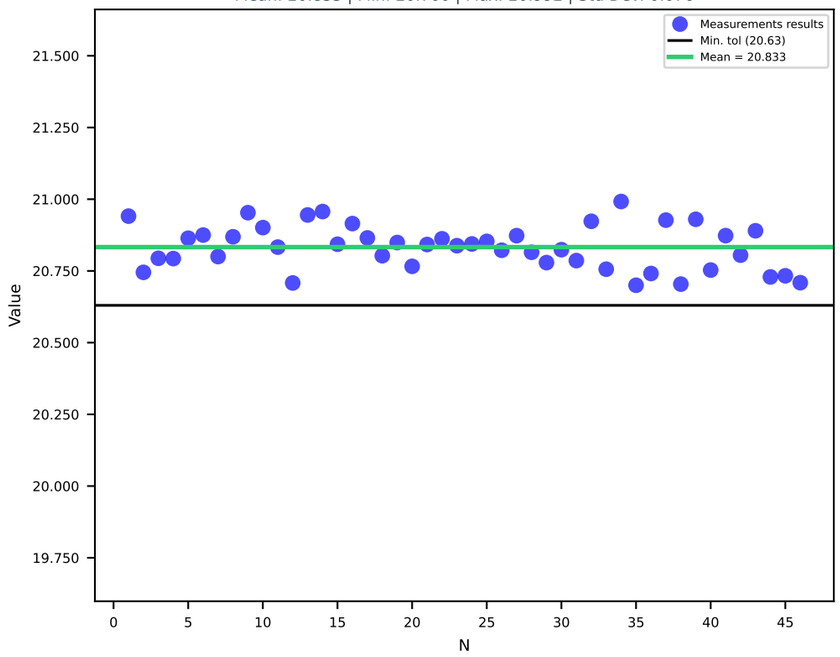


Technical Drawing:

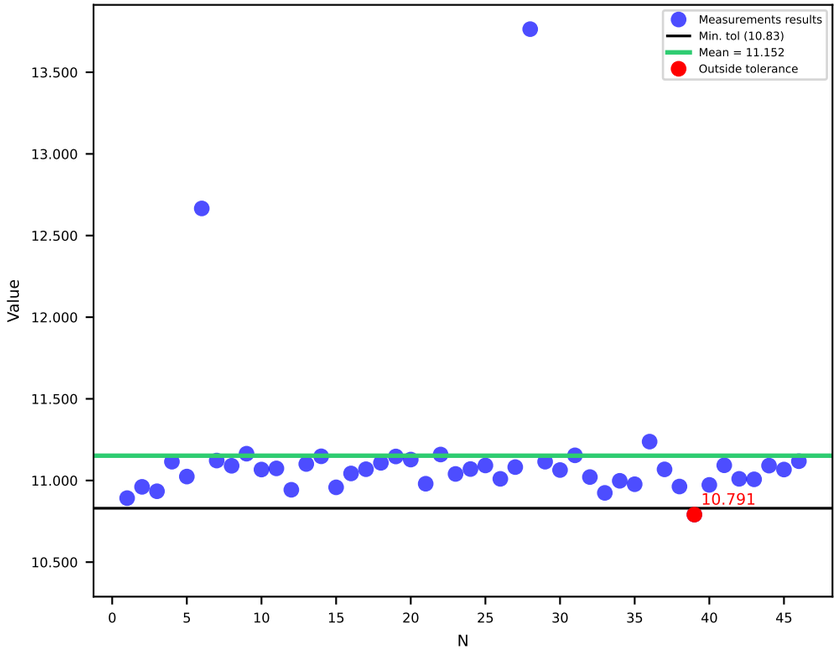
Technical drawing: DETAL-B



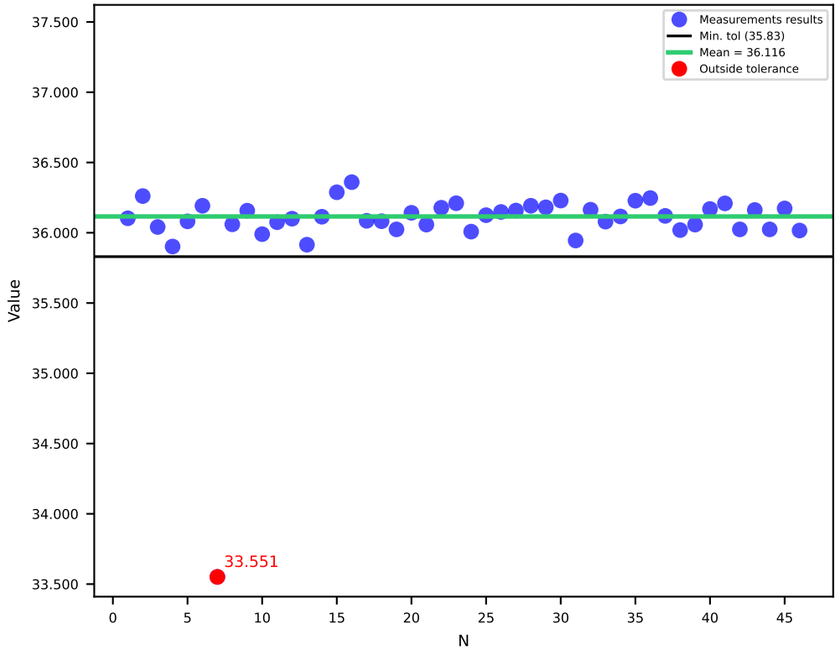
Char: 502 | N: 46 | Out: 0 (0.0%)
Mean: 20.833 | Min: 20.700 | Max: 20.992 | Std Dev: 0.076



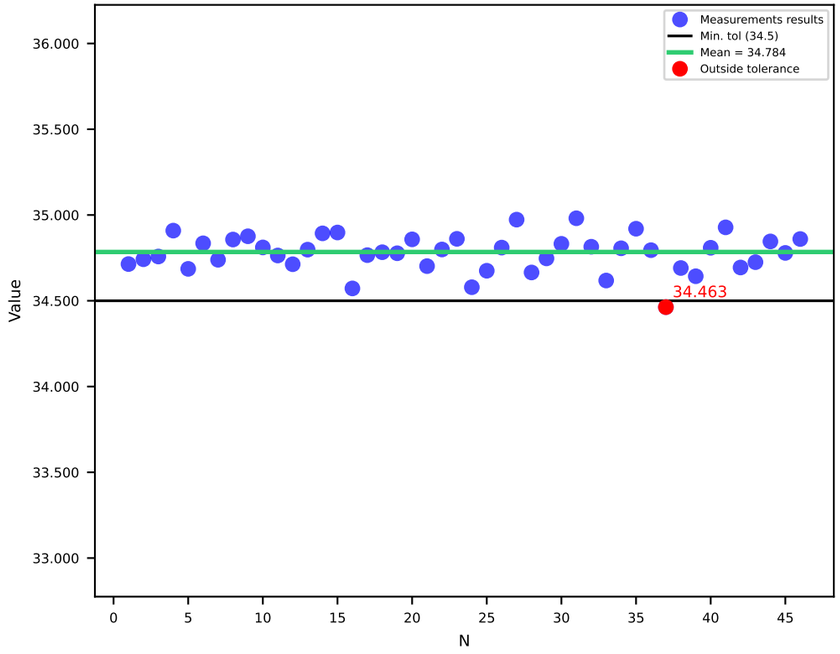
Char: 503 | N: 46 | Out: 1 (2.2%)
Mean: 11.152 | Min: 10.791 | Max: 13.764 | Std Dev: 0.471



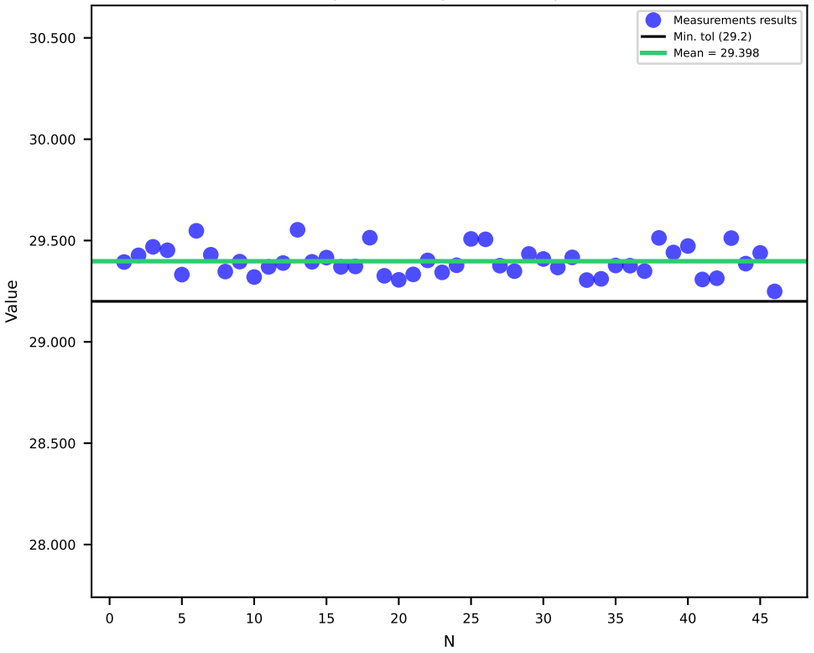
Char: 504 | N: 46 | Out: 1 (2.2%)
Mean: 36.116 | Min: 33.551 | Max: 36.360 | Std Dev: 0.097



Char: 505 | N: 46 | Out: 1 (2.2%)
Mean: 34.784 | Min: 34.463 | Max: 34.981 | Std Dev: 0.097

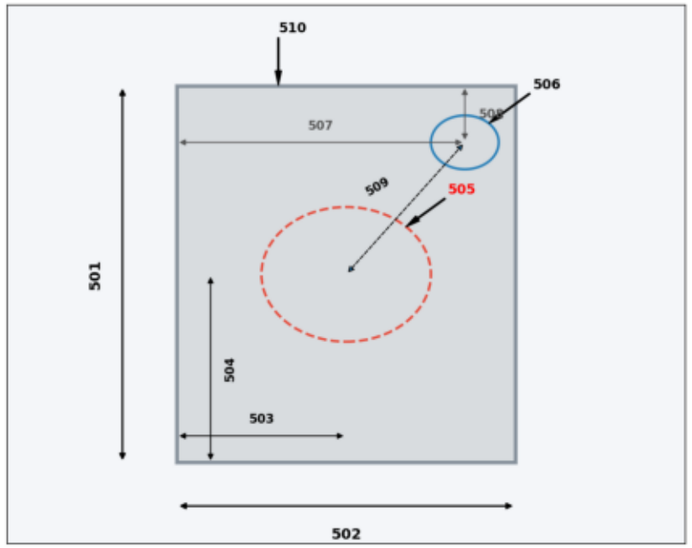


Char: 506 | N: 46 | Out: 0 (0.0%)
Mean: 29.398 | Min: 29.249 | Max: 29.553 | Std Dev: 0.071

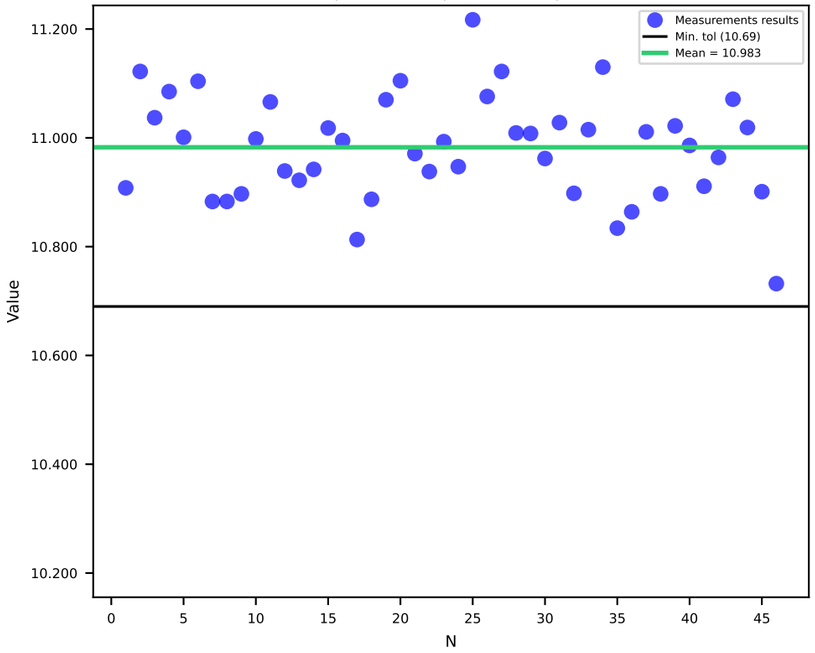


Technical Drawing:

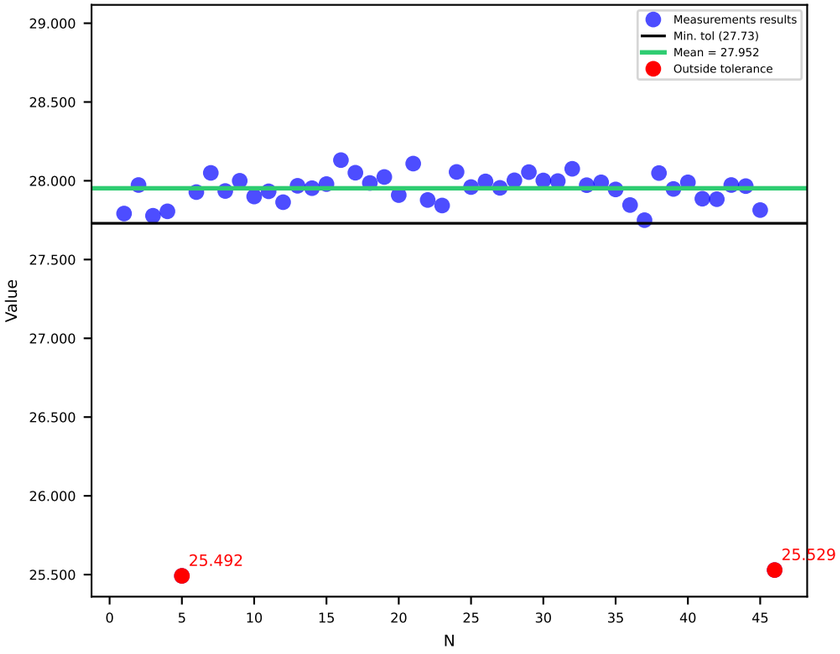
Technical drawing: DETAL-B



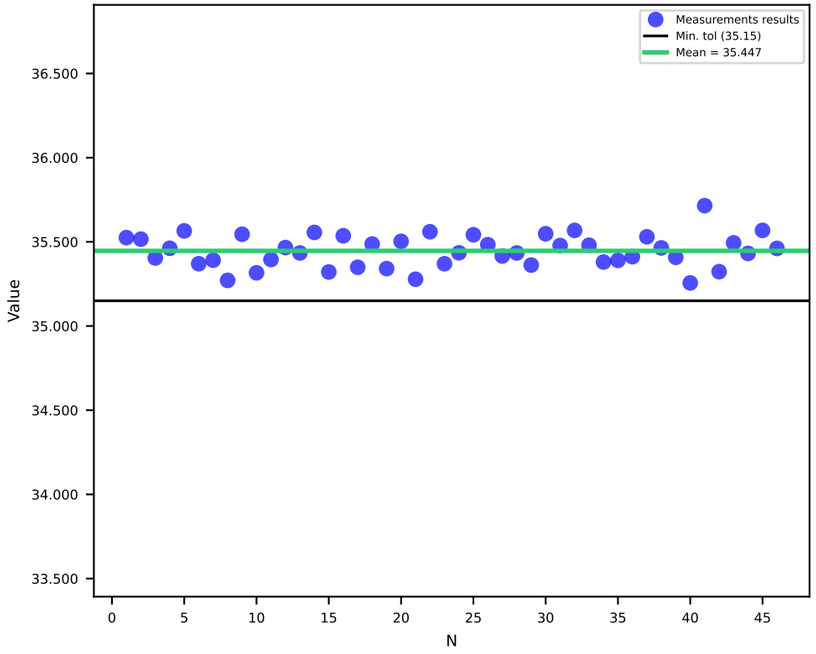
Char: 507 | N: 46 | Out: 0 (0.0%)
Mean: 10.983 | Min: 10.732 | Max: 11.217 | Std Dev: 0.095



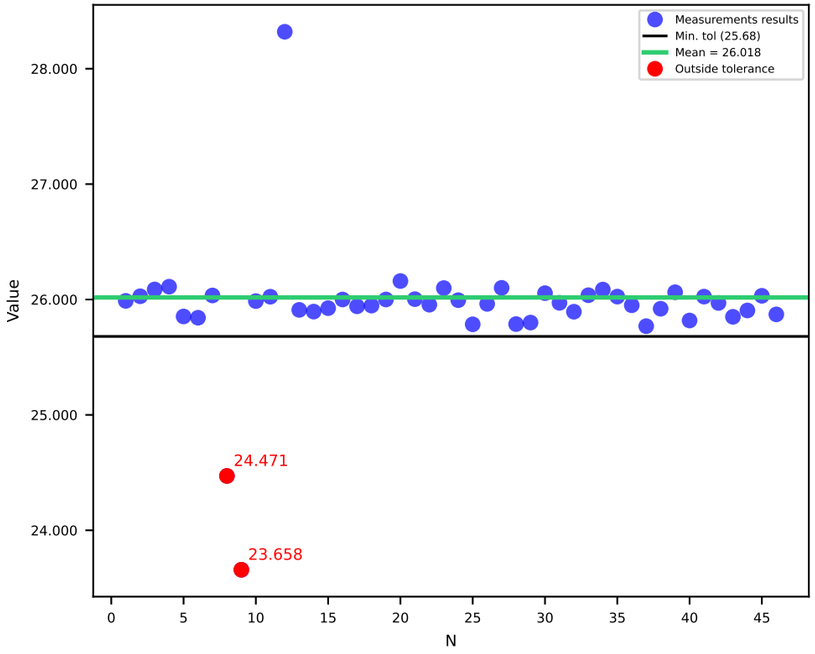
Char: 508 | N: 46 | Out: 2 (4.3%)
Mean: 27.952 | Min: 25.492 | Max: 28.131 | Std Dev: 0.088



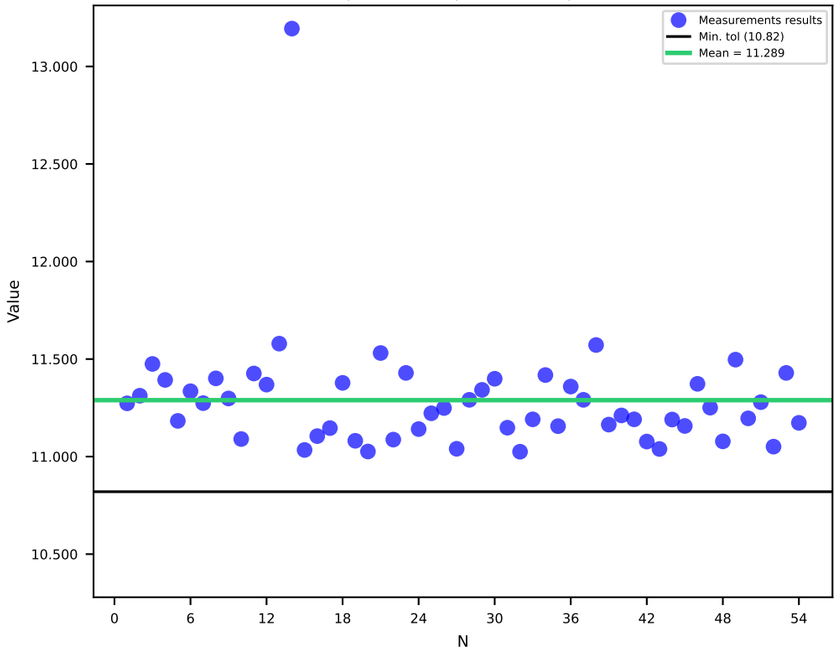
Char: 509 | N: 46 | Out: 0 (0.0%)
Mean: 35.447 | Min: 35.256 | Max: 35.715 | Std Dev: 0.095



Char: 510 | N: 46 | Out: 2 (4.3%)
Mean: 26.018 | Min: 23.658 | Max: 28.320 | Std Dev: 0.368

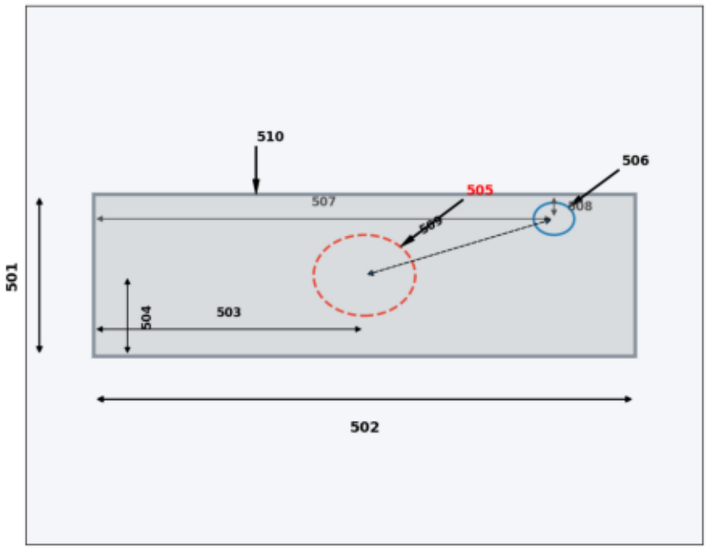


Char: 501 | N: 54 | Out: 0 (0.0%)
Mean: 11.289 | Min: 11.025 | Max: 13.194 | Std Dev: 0.304

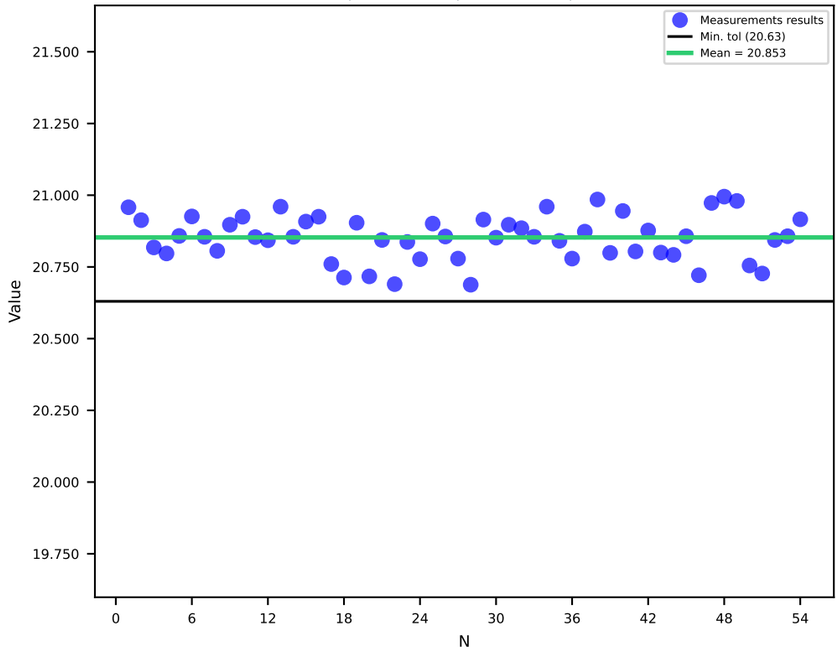


Technical Drawing:

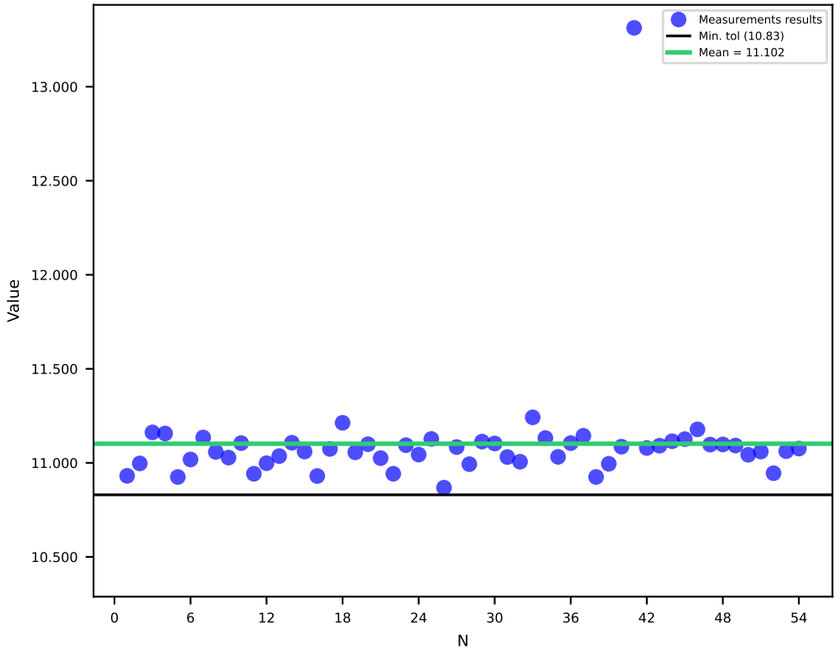
Technical drawing: DETAL-C



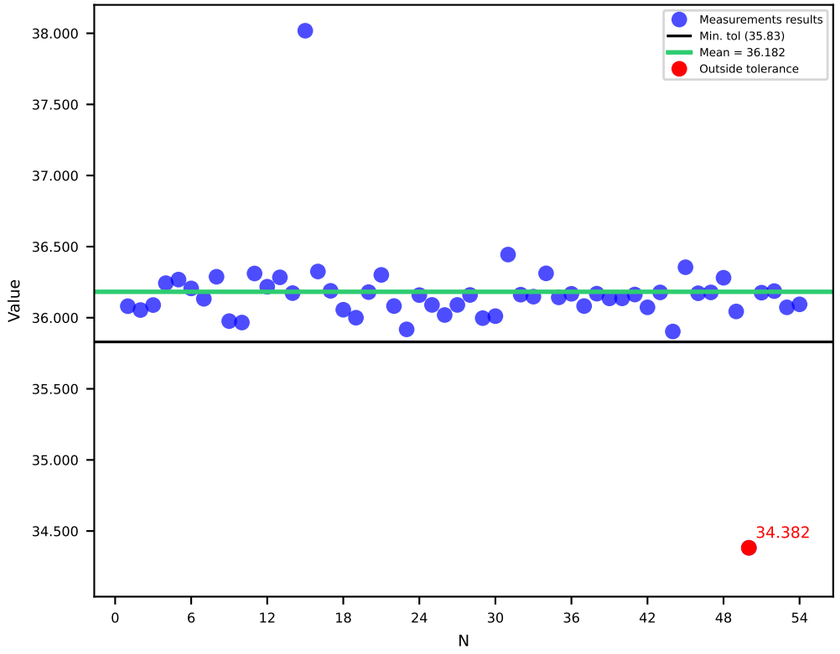
Char: 502 | N: 54 | Out: 0 (0.0%)
Mean: 20.853 | Min: 20.688 | Max: 20.995 | Std Dev: 0.079



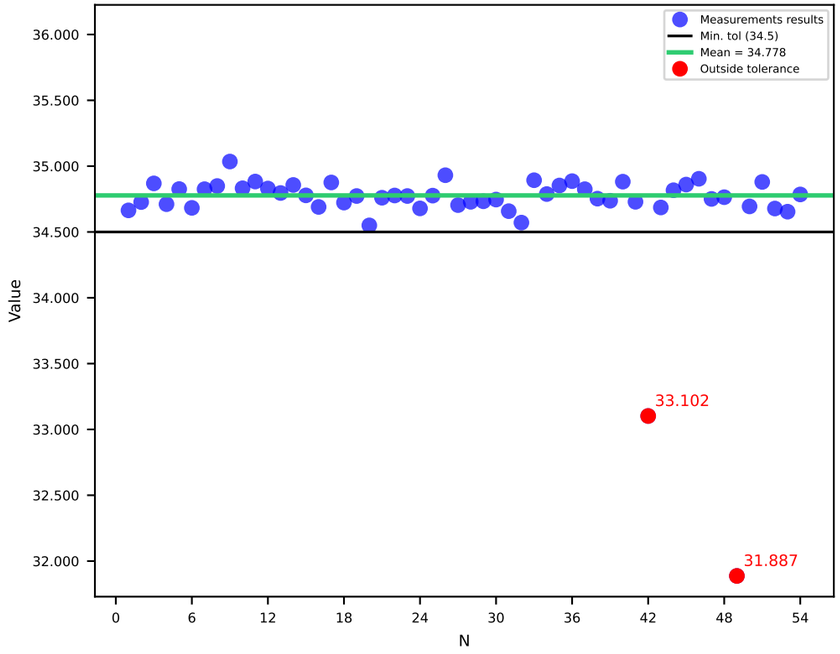
Char: 503 | N: 54 | Out: 0 (0.0%)
Mean: 11.102 | Min: 10.868 | Max: 13.313 | Std Dev: 0.316



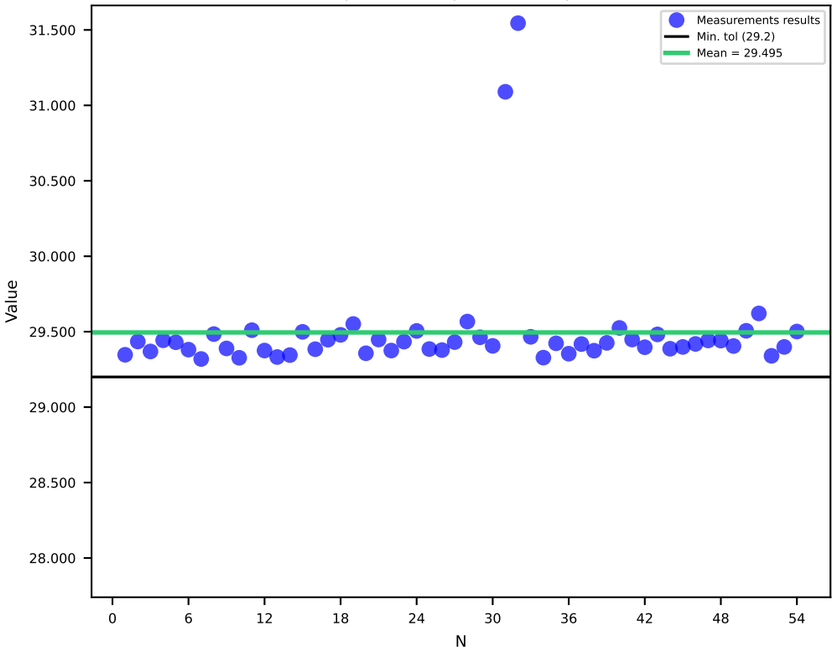
Char: 504 | N: 54 | Out: 1 (1.9%)
Mean: 36.182 | Min: 34.382 | Max: 38.018 | Std Dev: 0.281



Char: 505 | N: 54 | Out: 2 (3.7%)
Mean: 34.778 | Min: 31.887 | Max: 35.035 | Std Dev: 0.092

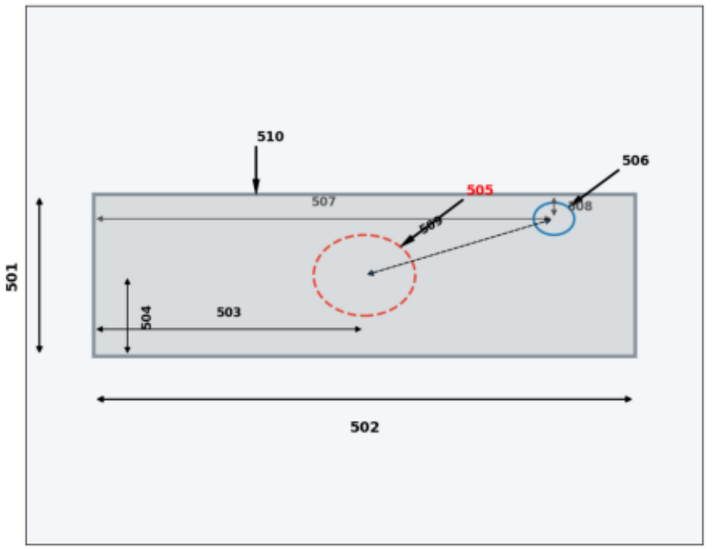


Char: 506 | N: 54 | Out: 0 (0.0%)
Mean: 29.495 | Min: 29.319 | Max: 31.545 | Std Dev: 0.369

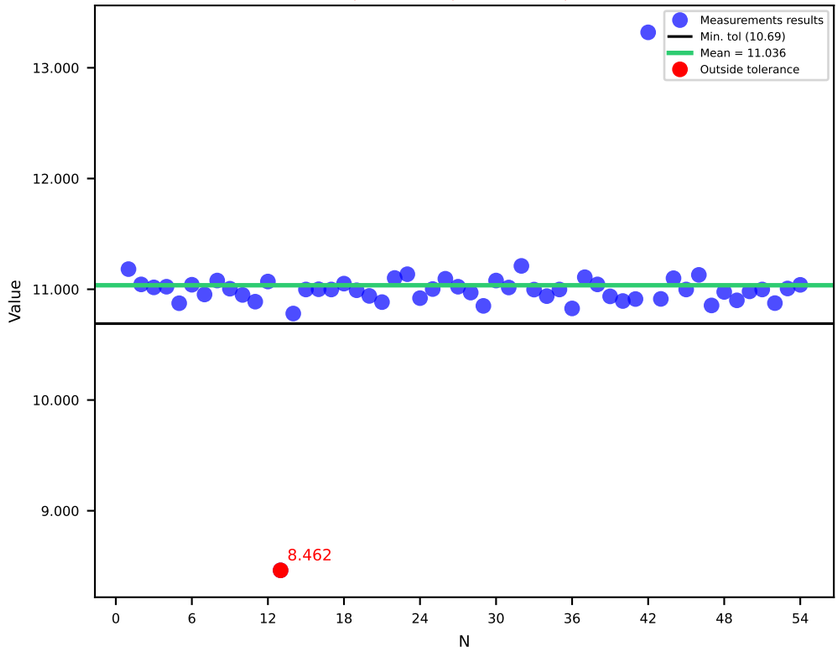


Technical Drawing:

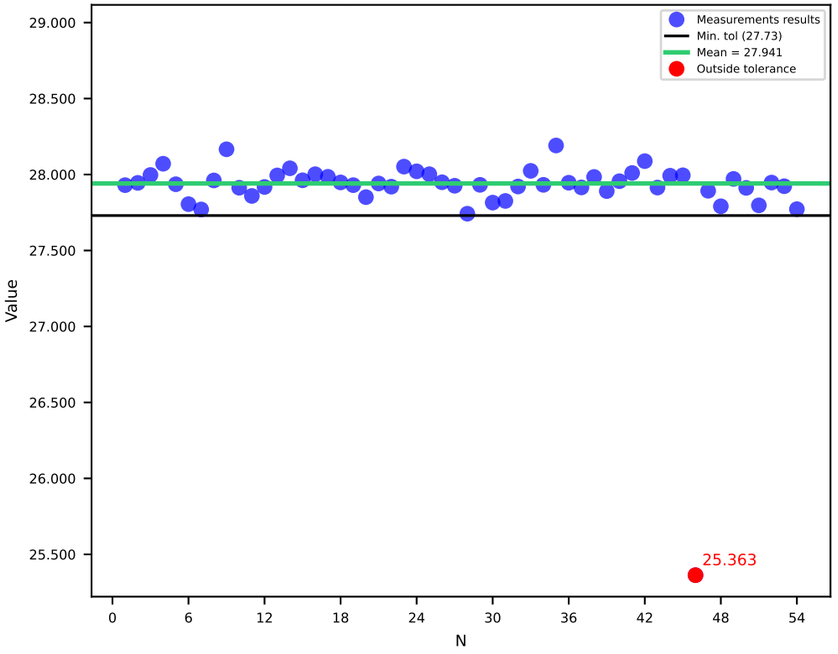
Technical drawing: DETAL-C



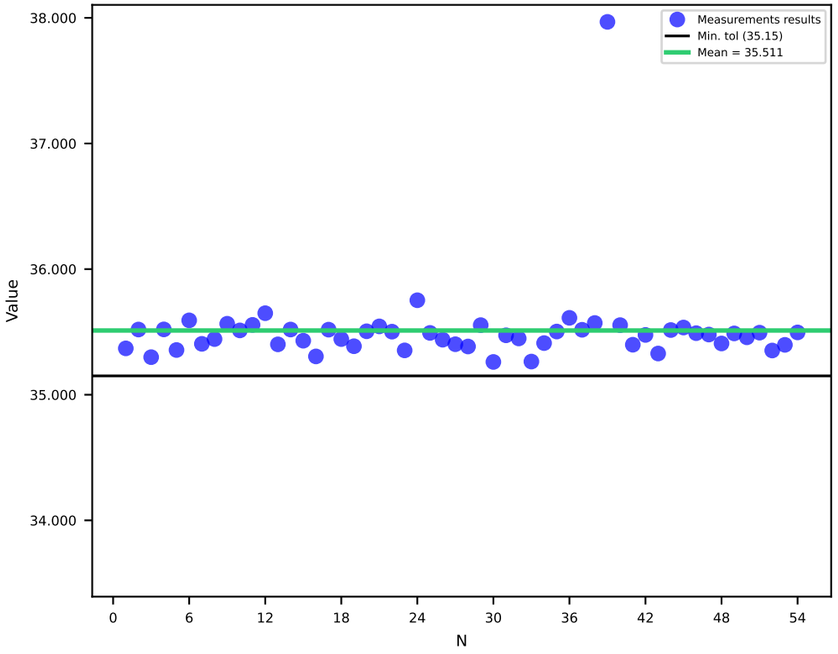
Char: 507 | N: 54 | Out: 1 (1.9%)
Mean: 11.036 | Min: 8.462 | Max: 13.320 | Std Dev: 0.332



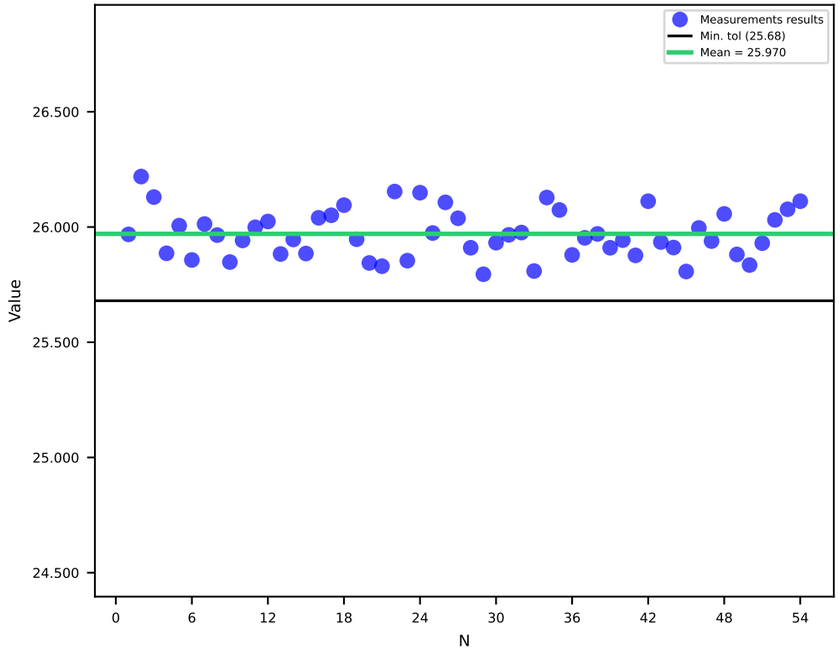
Char: 508 | N: 54 | Out: 1 (1.9%)
Mean: 27.941 | Min: 25.363 | Max: 28.191 | Std Dev: 0.091



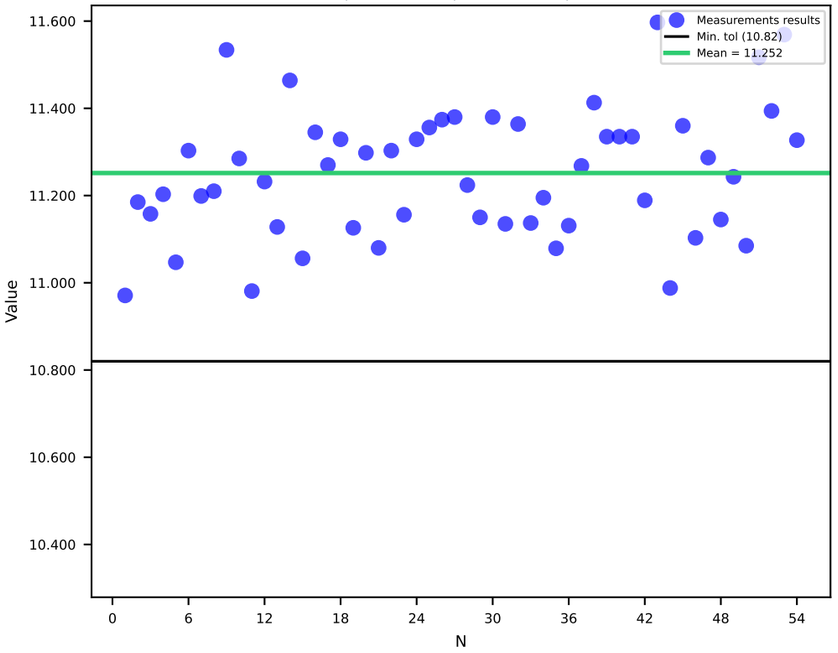
Char: 509 | N: 54 | Out: 0 (0.0%)
Mean: 35.511 | Min: 35.261 | Max: 37.968 | Std Dev: 0.354



Char: 510 | N: 54 | Out: 0 (0.0%)
Mean: 25.970 | Min: 25.795 | Max: 26.219 | Std Dev: 0.102

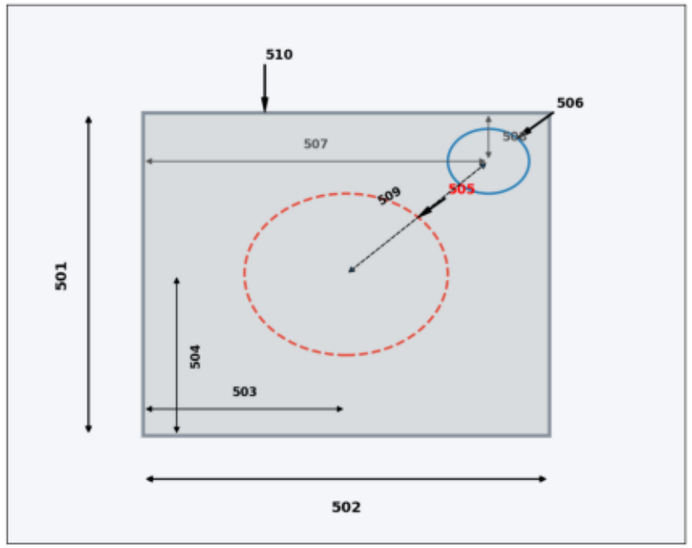


Char: 501 | N: 54 | Out: 0 (0.0%)
Mean: 11.252 | Min: 10.971 | Max: 11.597 | Std Dev: 0.147

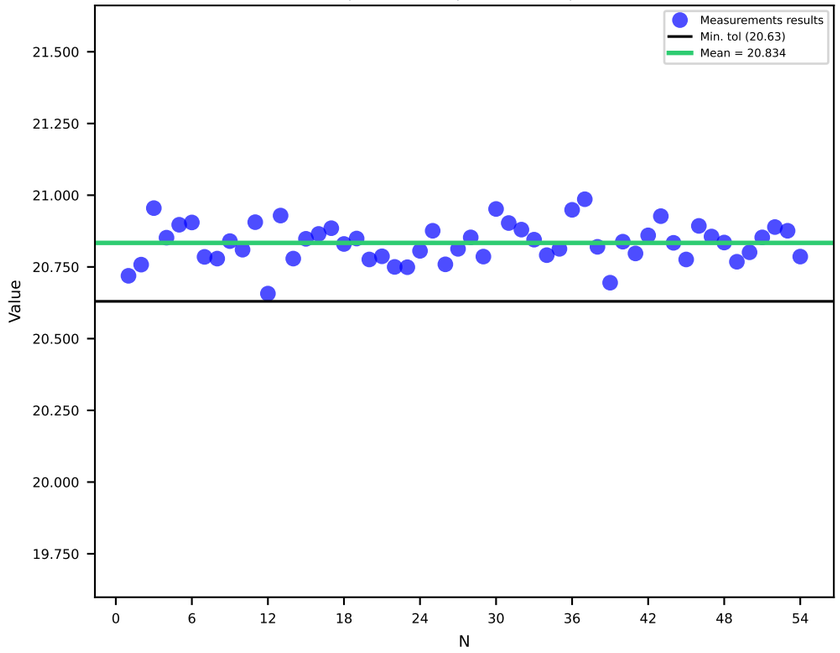


Technical Drawing:

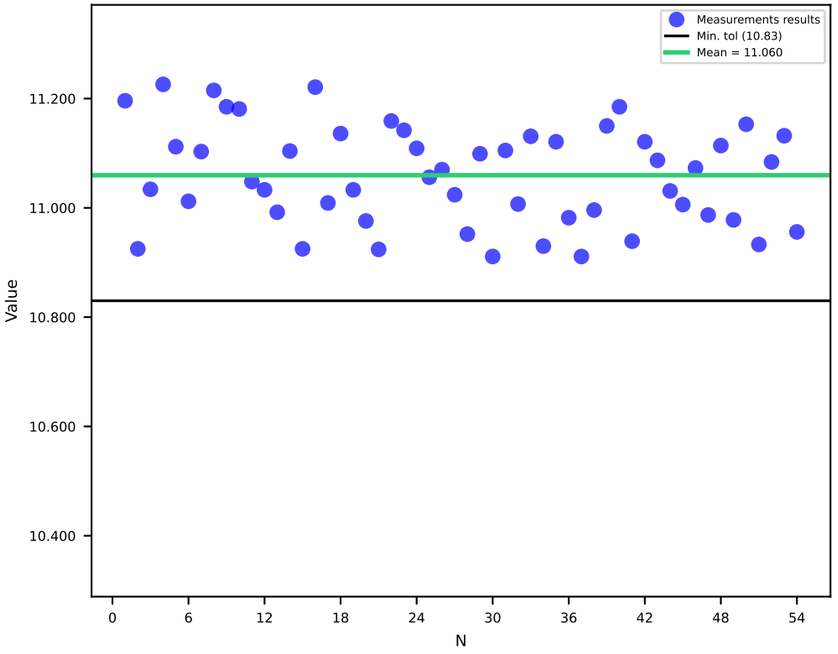
Technical drawing: DETAL-D



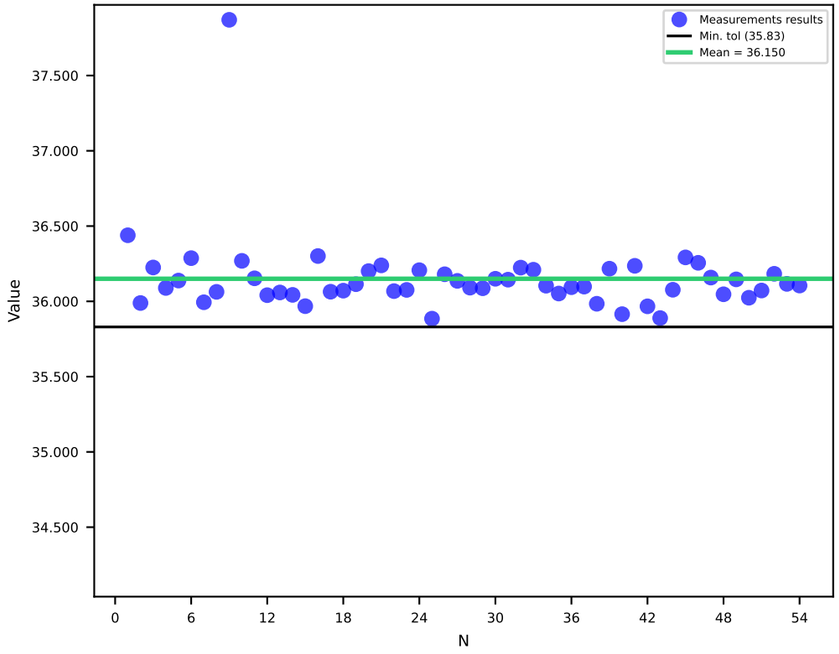
Char: 502 | N: 54 | Out: 0 (0.0%)
Mean: 20.834 | Min: 20.657 | Max: 20.986 | Std Dev: 0.068



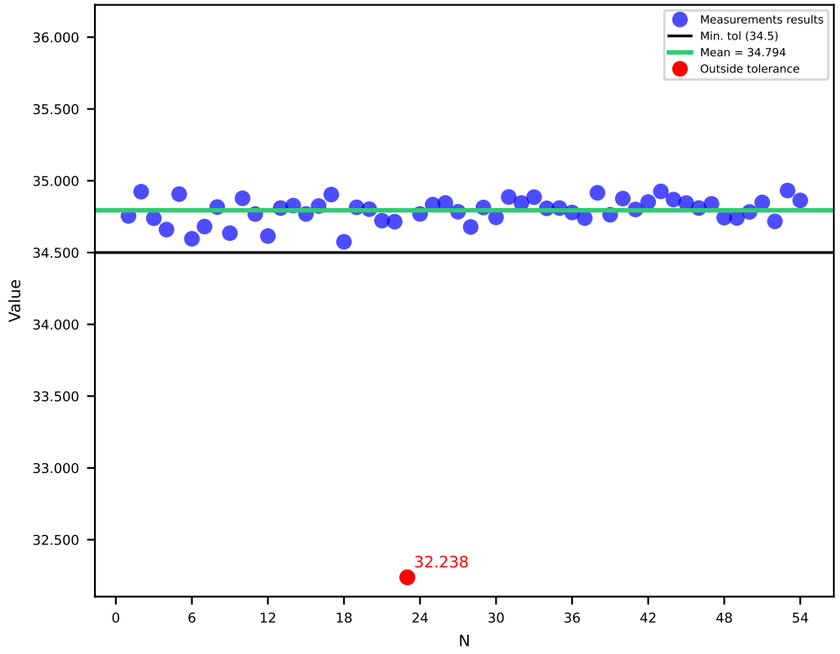
Char: 503 | N: 54 | Out: 0 (0.0%)
Mean: 11.060 | Min: 10.911 | Max: 11.226 | Std Dev: 0.090



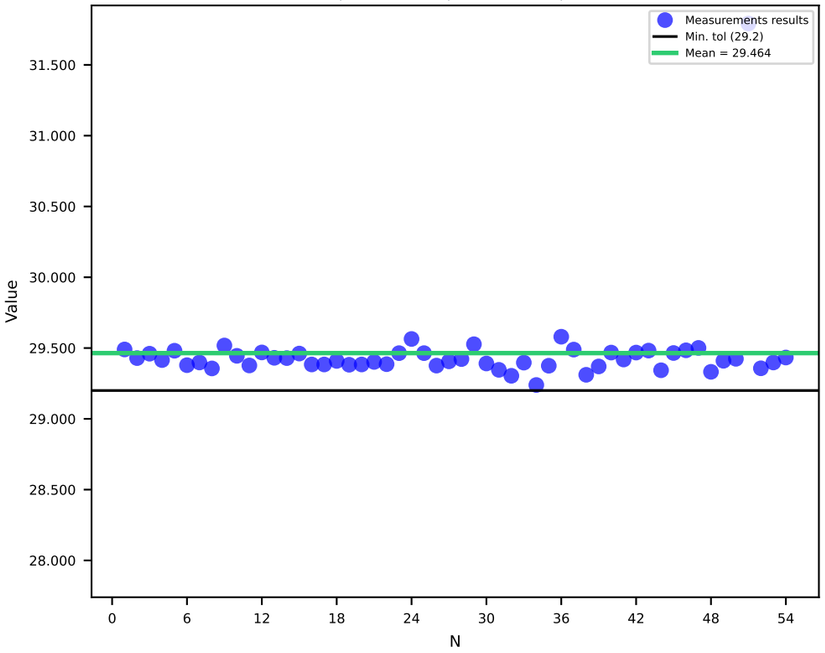
Char: 504 | N: 54 | Out: 0 (0.0%)
Mean: 36.150 | Min: 35.885 | Max: 37.870 | Std Dev: 0.262



Char: 505 | N: 54 | Out: 1 (1.9%)
Mean: 34.794 | Min: 32.238 | Max: 34.932 | Std Dev: 0.086

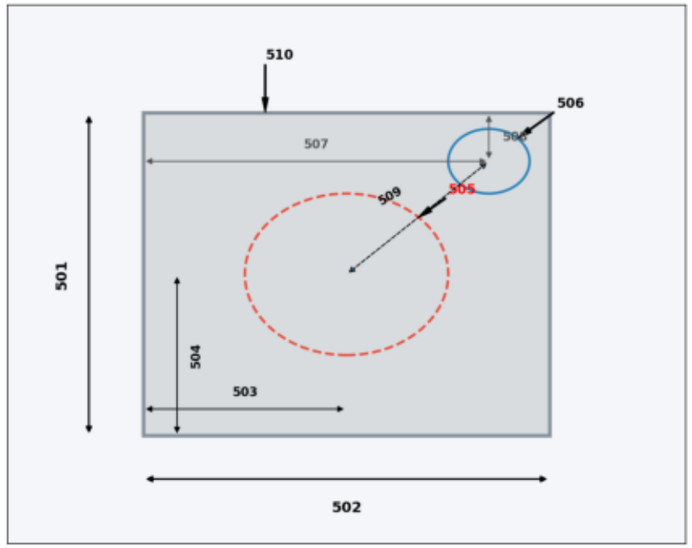


Char: 506 | N: 54 | Out: 0 (0.0%)
Mean: 29.464 | Min: 29.239 | Max: 31.792 | Std Dev: 0.329

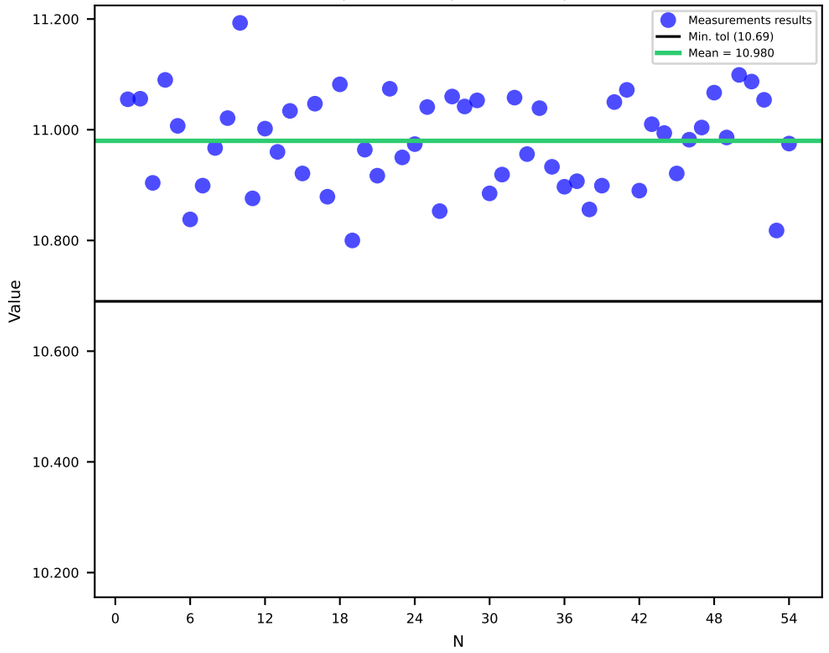


Technical Drawing:

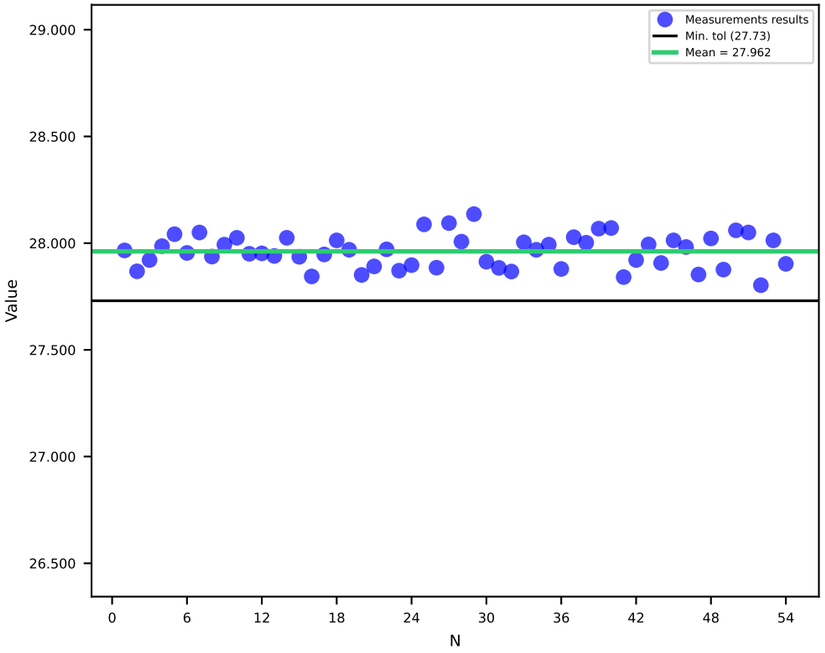
Technical drawing: DETAL-D



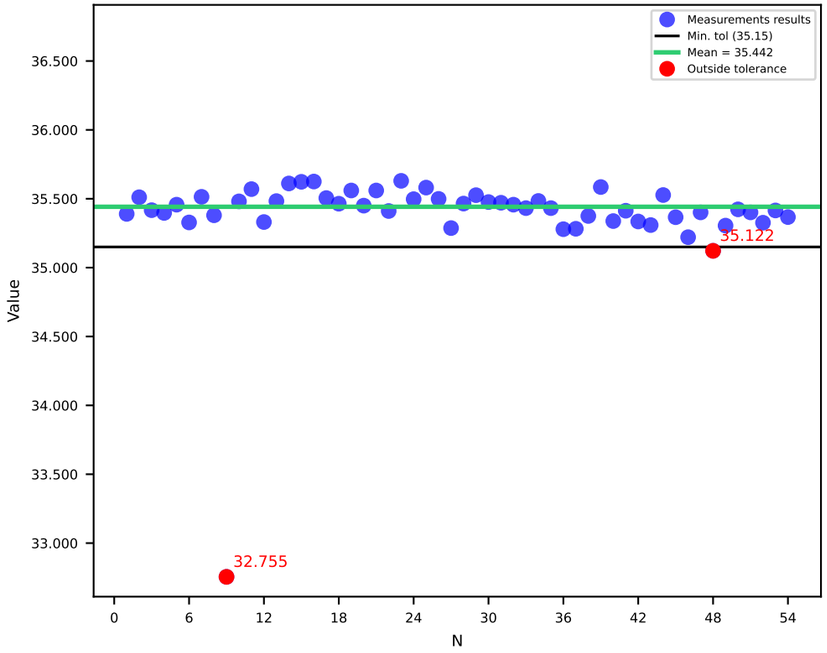
Char: 507 | N: 54 | Out: 0 (0.0%)
Mean: 10.980 | Min: 10.800 | Max: 11.193 | Std Dev: 0.085



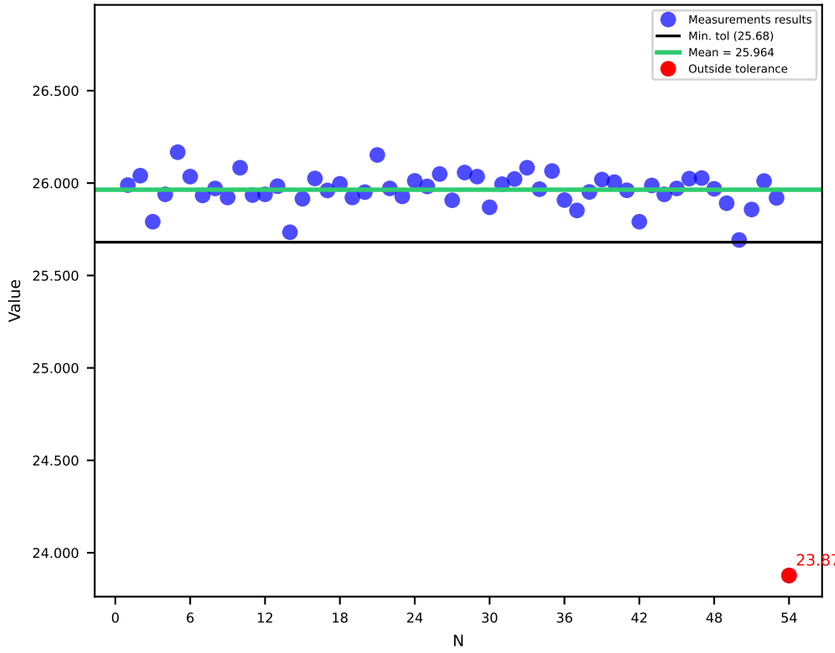
Char: 508 | N: 54 | Out: 0 (0.0%)
Mean: 27.962 | Min: 27.803 | Max: 28.136 | Std Dev: 0.076



Char: 509 | N: 54 | Out: 2 (3.7%)
Mean: 35.442 | Min: 32.755 | Max: 35.630 | Std Dev: 0.100



Char: 510 | N: 54 | Out: 1 (1.9%)
Mean: 25.964 | Min: 23.877 | Max: 26.167 | Std Dev: 0.090



Outliers summary

Part ID	Characteristic	Measured	Min. Tol.	Deviation
DETAL-A	501	9.723	10.820	-1.097
DETAL-A	501	10.789	10.820	-0.031
DETAL-A	502	17.962	20.630	-2.668
DETAL-A	502	18.812	20.630	-1.818
DETAL-A	502	19.032	20.630	-1.598
DETAL-A	502	19.345	20.630	-1.285
DETAL-A	507	9.216	10.690	-1.474
DETAL-A	507	8.435	10.690	-2.255
DETAL-A	509	33.065	35.150	-2.085
DETAL-B	501	9.366	10.820	-1.454
DETAL-B	503	10.791	10.830	-0.039
DETAL-B	504	33.551	35.830	-2.279
DETAL-B	505	34.463	34.500	-0.037
DETAL-B	508	25.492	27.730	-2.238
DETAL-B	508	25.529	27.730	-2.201
DETAL-B	510	24.471	25.680	-1.209
DETAL-B	510	23.658	25.680	-2.022
DETAL-C	504	34.382	35.830	-1.448
DETAL-C	505	33.102	34.500	-1.398
DETAL-C	505	31.887	34.500	-2.613
DETAL-C	507	8.462	10.690	-2.228
DETAL-C	508	25.363	27.730	-2.367
DETAL-D	505	32.238	34.500	-2.262
DETAL-D	509	32.755	35.150	-2.395
DETAL-D	509	35.122	35.150	-0.028

Summary continuation

Part ID	Characteristic	Measured	Min. Tol.	Deviation
DETAL-D	510	23.877	25.680	-1.803

End of Report

All measurements processed and validated successfully.